IRRATIONALITY MEASURES FOR $G \pm 5\pi^2/48$, $L(2,\chi_{-3})$, $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$ AND $\psi'(\frac{a}{5}) - \zeta(2)$

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ABSTRACT. Building on OpenAI's determinant method for the irrationality of Catalan's constant $G = \sum_{j\ge 0}(-1)^j/(2j+1)^2$, we prove

$$\mu\Big(G - \frac{5\pi^2}{48}\Big) \le 32.44, \qquad \mu\Big(G + \frac{5\pi^2}{48}\Big) \le 32.44,$$
$$\mu\big(L(2,\chi_{-3})\big) \le 14.21,$$

and, for the digamma differences with shifts $\frac{1}{3}$ and $\frac{2}{3}$, $\mu\big(L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)\big) \le 10.61$, where μ is the irrationality exponent. We know of no earlier irrationality measure for $G \pm 5\pi^2/48$. For $L(2,\chi_{-3})$, which Calegari, Dimitrov and Tang proved irrational, the previous best bound we know of is their effective bound $\mu \le 24781$. The same method gives, for the two numbers $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$,

$$\mu\Big(L(2,\chi_{-3}) \pm \tfrac{2}{3}\zeta(2)\Big) \le 10.61.$$

For the shifts $\frac{a}{5}$ it gives $\mu\big(\psi'(\frac{a}{5}) - \zeta(2)\big) \le 31.95$ for $a = 1,2,3,4$; we know of no earlier irrationality result for these four numbers.

The approximations come from determinants of *rectangular* Löwner matrices of digamma differences at integer nodes. We bound their denominators with p-adic assignment problems and rank deficits modulo p. We prove nonvanishing modulo a prime by reducing it to bordered Krylov determinants, which we evaluate through Jacobi matrices and Darboux transformations. For $L(2,\chi_{-3})$, a finite-field interpolation argument bounds the corank by three; removing at most three common low nodes gives nonvanishing at a fixed scale throughout a prime interval. We bound their size with Andréief's identity and a certified lower bound for a discrete logarithmic energy. Two new ingredients lower the exponent for $G \pm 5\pi^2/48$: a second-level p-adic saving at primes where the leading block is not clean, and a decay certificate whose atoms are placed on the support of an extremal dual measure, with steeper contour rays.

Every result is formalized in Lean 4, with no unproved steps and only the standard axioms.

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1. Introduction

In October 2026 OpenAI [36] proved that Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = 0.91596\,55941\ldots$$

is irrational, and formalized the proof in Lean. (An earlier claimed proof [44] was found to contain a gap [47].) Their argument, and the irrationality-measure results built on it [20, 9, 29], use a determinant of large size whose denominators and decay are balanced very closely.

In the present paper we study a different, much smaller, family of determinants: rectangular Löwner matrices of digamma differences, attached to the numbers

$$\lambda_- = \frac{G}{2} - \frac{5\pi^2}{96}, \qquad \lambda_+ = \frac{G}{2} + \frac{5\pi^2}{96}, \qquad \frac{1}{4}L(2,\chi_{-3}),$$
$$L(2,\chi_{-3}) = \sum_{k\ge 0}\Big(\frac{1}{(3k+1)^2} - \frac{1}{(3k+2)^2}\Big) = 0.78130\,24128\ldots$$

Thus $G - 5\pi^2/48 = 2\lambda_-$, $G + 5\pi^2/48 = 2\lambda_+$, and $\lambda_- + \lambda_+ = G$.

Recall that the *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large b.

Theorem 1.1. *For every $\varepsilon > 0$ there is q_0 such that for all coprime integers p, q with $q \geq q_0$,*

$$\left|G - \frac{5\pi^2}{48} - \frac{p}{q}\right| > q^{-32.44-\varepsilon}, \qquad \left|G + \frac{5\pi^2}{48} - \frac{p}{q}\right| > q^{-32.44-\varepsilon},$$
$$\left|L(2,\chi_{-3}) - \frac{p}{q}\right| > q^{-14.21-\varepsilon}.$$

In particular $G \pm 5\pi^2/48$ are irrational, $\mu(G \pm 5\pi^2/48) \leq 32.44$ and $\mu(L(2,\chi_{-3})) \leq 14.21$.

The computed values of the exponents are 32.4392 and $14.2095662593\ldots$.

A companion result concerns the digamma differences with shifts $\frac{1}{3}$ and $\frac{2}{3}$, for which $\psi'(\frac{1}{3}) = \frac{2\pi^2}{3} + \frac{9}{2}L(2,\chi_{-3})$ and $\psi'(\frac{2}{3}) = \frac{2\pi^2}{3} - \frac{9}{2}L(2,\chi_{-3})$. Since $\pi^2/9 = \frac{2}{3}\zeta(2)$, they give the numbers $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$.

Theorem 1.2. *For every $\varepsilon > 0$ there is q_0 such that for all coprime integers p, q with $q \geq q_0$,*

$$\left|L(2,\chi_{-3}) + \frac{\pi^2}{9} - \frac{p}{q}\right| > q^{-10.61-\varepsilon}, \qquad \left|L(2,\chi_{-3}) - \frac{\pi^2}{9} - \frac{p}{q}\right| > q^{-10.61-\varepsilon}.$$

In particular $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$ are irrational and have irrationality exponent at most 10.61.

The computed value of the exponent is $10.6078\ldots$, namely $6.78788/(3.87115-3.2312582\ldots)$.

The shifts $\frac{a}{5}$ give four further numbers, $\psi'(\frac{a}{5}) - \zeta(2) = \sum_{n\geq 0}(n+\frac{a}{5})^{-2} - \frac{\pi^2}{6}$.

Theorem 1.3. *For $a \in \{1,2,3,4\}$ and every $\varepsilon > 0$ there is q_0 such that for all coprime integers p, q with $q \geq q_0$,*

$$\left|\psi'(\tfrac{a}{5}) - \zeta(2) - \frac{p}{q}\right| > q^{-31.95-\varepsilon}.$$

In particular the four numbers $\psi'(\frac{a}{5}) - \zeta(2)$ are irrational and have irrationality exponent at most 31.95.

The computed value of the exponent is $31.9439\ldots$. Theorems 1.1, 1.2 and 1.3, together with every result in this paper, has been verified in the Lean 4 proof assistant, with no unproved steps and only the standard axioms; see Section 14. The formalization is available at `https://github.com/slitsey/catalan-irrationality-measure`. The use of AI in this work is disclosed at the end of the paper.

1.1. **Previous results.** For earlier work on the arithmetic of G and related values see [41, 51, 50, 25, 26] and the survey [53]; for recent rational approximations to G see Eskandari [12]. Finite irrationality exponents for G itself were obtained after OpenAI's proof by Jha [20], Carreira [9] and the author [29]; Carreira [8] also records attempts to carry the determinant method to $\beta(4)$. Our searches found no previous irrationality result, and no irrationality measure, for $G - 5\pi^2/48$ or $G + 5\pi^2/48$. Eskandari, Murty and Nemoto

[13] construct a 2-dimensional mixed motive over $\mathbb{Q}$ with period G and use it to obtain linear forms in 1 and G. Eskandari [12] improves the approximation exponent of these Rivoal–Zudilin type forms from 0.52 to 0.62 (a preprint; explicit approximations, not a lower bound for all rationals). Sun's preprint [44] claimed a proof of the irrationality of G; Wachs [47] reports that its main estimate is not justified.

Calegari, Dimitrov and Tang proved that 1, $\zeta(2)$ and $L(2,\chi_{-3})$ are linearly independent over $\mathbb{Q}$ [6], by a new arithmetic holonomy bound applied to a construction of Zagier. In [7] they made holonomy bounds effective; Theorem 5.1 there gives $\mu(L(2,\chi_{-3})) \le 24781$. The remark following that theorem states that the same theorem and proof hold for any $\mathbb{Q}$-linear combination of $L(2,\chi_{-3})$ and $\zeta(2)$; in particular 24781 was the known bound for $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$. Theorem 1.1 improves it to 14.21 for $L(2,\chi_{-3})$, and Theorem 1.2 to 10.61 for the two numbers $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$. Leibtag et al. [28] find numerically that Apéry-like approximations from conservative matrix fields have positive approximation exponents for $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$, and a supplement to their paper uses one such field to give an Apéry-style proof of the theorem of Calegari, Dimitrov and Tang; the exponent it proves for each of these two numbers is only greater than 10^{-4}. For the four numbers $\psi'(\frac{a}{5}) - \zeta(2)$ of Theorem 1.3 our searches found no earlier irrationality result; the irrationality of $\mathrm{Li}_2(1/5)$ proved by Prellberg [38] concerns a different number.

For comparison, among classical constants, the bound $\mu(\log 2) < 3.5746$ of Marcovecchio [33] and the Rhin–Viola bounds for $\zeta(2)$ and $\zeta(3)$ [39, 40] (after Apéry and Beukers [2, 4]) have been joined, since September 2026, by a series of new results, several of them obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [37], improving the bound $\mu(\pi) < 7.1033$ of Zeilberger and Zudilin [48]. Liu, Jiang and Zhang proved $\mu(\log a) = 2$ for every rational $a > 0$, $a \neq 1$, and for nonzero rational combinations of such logarithms [30]. Giraudin [16, 17] lowered the bounds for $\mu(\zeta(2))$ and $\mu(\zeta(3))$ to 5.0193784 and 5.5138800 by applying the Rhin–Viola method at new integral directions. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [28] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields, and by searching over other trajectories obtained $\mu(\zeta(2)) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [38] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational, three cases left open in [6], with the integrals of Rhin–Viola and Viola–Zudilin. Fauzan [14] announced a proof that $\zeta(5)$ is irrational, using rationally normalized Hankel determinants, and Lai [27] combined his method with simultaneous approximations of Zudilin and of Dauguet and Zudilin to show that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$.

Related work on determinants. Irrationality proofs built on determinants of large size have a precedent in Zudilin's determinantal criterion with Hankel determinants [52]; the preprint of Liu, Jiang and Zhang on logarithms [30] uses a separated-weight interpolation determinant, and the arguments of

Fauzan and Lai [14, 27] use Hankel determinants. Brown [5] proposes a multivariate transfinite-diameter test for Mellin-type integrals, applied to $\zeta(2)$. The framework we use, determinants whose decay is controlled by Cauchy determinants and Andréief's identity and whose denominators are controlled prime by prime, is OpenAI's [36]. Our determinants are Löwner (divided-difference) matrices of a digamma difference, a matrix-valued analogue of Padé and Hermite–Padé approximation [35]. Formal proofs of irrationality results in Lean also include the irrationality of $\zeta(3)$ by Liu, Zhang and Zhi [31].

1.2. **Outline of the proof.** Section 2 explains the construction and the new ideas without technicalities, and states the general deduction. Sections 3–9 give the construction and the three estimates (denominator, nonvanishing, decay); Section 10 deduces Theorem 1.1, and Section 11 proves Theorem 1.2, Section 12 proves Theorem 1.3, and Section 13 discusses the limits of the method. Section 14 describes the formalization, and the disclosure of AI use follows it.

1.3. **Notation and conventions.** $\psi = \Gamma'/\Gamma$ is the digamma function; v_p is the p-adic valuation; for $P \in \mathbb{Q}[x]$, $\operatorname{den} P$ is the least positive integer d with $dP \in \mathbb{Z}[x]$; $\vartheta(x; M, a) = \sum_{p \le x,\ p \equiv a\ (M)} \log p$; $(x)_n = x(x+1)\cdots(x+n-1)$; $[0, r) = \{0, 1, \dots, r-1\}$. For finite sets I, J of integers and a matrix A indexed by integers, $A[I, J]$ is the submatrix with rows I and columns J.

2. Overview of the method and of the new ideas

2.1. **Löwner matrices of digamma differences.** For a function F analytic near distinct points $x_0, \dots, x_{h-1}$ the *Löwner matrix* [32] is the matrix of divided differences

$$\mathsf{L}_F[a, b] = \frac{F(x_b) - F(x_a)}{x_b - x_a} \ (a \neq b), \qquad \mathsf{L}_F[a, a] = F'(x_a).$$

It is a Cauchy-like structured matrix, and Hermite's formula gives $\mathsf{L}_F = \frac{1}{2\pi i} \oint F(z)\, v(z) v(z)^{\top} dz$ with $v_a(z) = 1/(z - x_a)$, so that a representation of F as a Stieltjes or contour integral exhibits L_F as a Gram matrix of Cauchy kernels.

We take three digamma differences, with integer nodes:

$$\begin{gathered} F_{\lambda_-}(z) = \psi(z+1) - \psi(z+\tfrac{3}{4}), \quad \omega = \tfrac{1}{16}, \quad \xi = \lambda_- := \tfrac{G}{2} - \tfrac{5\pi^2}{96}, \\ F_{\lambda_+}(z) = \psi(z+1) - \psi(z+\tfrac{1}{4}), \quad \omega = -\tfrac{1}{16}, \quad \xi = \lambda_+ := \tfrac{G}{2} + \tfrac{5\pi^2}{96}, \\ F_{\chi}(z) = -\tfrac{1}{3}\psi(z+\tfrac{1}{3}) + \tfrac{1}{2}\psi(z+\tfrac{1}{2}) - \tfrac{1}{6}\psi(z+1), \\ \omega = -\tfrac{1}{6}, \quad \xi = \tfrac{1}{4} L(2, \chi_{-3}). \end{gathered} \tag{2.1}$$

In each case $\xi = \omega\, F'(0)$ (this is an evaluation of ψ' at $\frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, 1$), while $F(b) - F(a)$ and $F'(a) - F'(0)$ are rational for integers $a, b \ge 0$. Hence, for

any finite sets of nonnegative integer nodes R (rows) and Q (columns),

$$\omega \llcorner_F [R,Q] = M[R,Q] + \xi\, E_{R,Q}, \qquad E_{R,Q}[a,b] = \mathbf{1}[a=b], \tag{2.2}$$

with an explicit rational symmetric kernel M on $\mathbb{Z}^2_{\geq 0}$ (§3). Replacing ξ by a variable x gives a matrix polynomial $\Phi(x) = M[R,Q] + xE_{R,Q}$, whose determinant $P(x)$ is a polynomial with rational coefficients of degree at most $|R \cap Q|$, and $P(\xi) = \det(\omega \llcorner_F [R,Q])$. If $|\xi - p/q|$ is very small, $P(p/q)$ is very close to $P(\xi)$, which is tiny; on the other hand $q^{|R\cap Q|}\operatorname{den}(P)\,P(p/q)$ is an integer. If it is not zero we get a contradiction as soon as

$$\underbrace{\log \operatorname{den} P}_{\text{content}} + |R \cap Q| \log q \;<\; \underbrace{-\log |P(p/q)|}_{\text{decay}}.$$

Three estimates are therefore needed: an upper bound for the *content* $\log \operatorname{den} P$, an upper bound for $|P(x)|$ near ξ (the *decay*), and *nonvanishing* of $P(p/q)$ for a size that can be chosen in terms of q. We use rectangular geometries, with $R \neq Q$, and all three estimates are new for these constants.

2.2. **New idea 1: rectangular frames.** We take a long common block C of r nodes, which carries the variable x, and two short interleaved blocks of $m \approx r/8$ nodes,

$$C = [0, r), \qquad X' = \{r, r+2, \ldots, r+2m-2\} \text{ (rows only)},$$
$$Y' = \{r+1, r+3, \ldots, r+2m-1\} \text{ (columns only)},$$

so that $R = C \sqcup X'$ and $Q = C \sqcup Y'$ (a second "member" adds one more common node $r+2m$). The determinant $P(x) = \det \Phi(x)$ still has degree $\leq r$ (or $r+1$), so the integrality loss $|R \cap Q| \log q$ is that of r nodes, but the matrix has size $r+m$ and its decay is that of a configuration of $(1+\alpha)r$ nodes, $\alpha = m/r$. The content grows too, but less: the extra rows and columns never meet the diagonal. With $\alpha = \frac{1}{8}$ the content and decay constants are (for λ_-)

$$\Lambda = 3.67885, \qquad D = 3.90557,$$

and the margin is $\eta = D - \Lambda = 0.22672$, larger than for the square matrix. The interleaving of X' and Y' is essential: it keeps the two Cauchy determinants in Andréief's formula comparable (§9) and it gives the reduction modulo ℓ the bordered structure of §7.

2.3. **New idea 2: content by p-adic assignment problems and rank deficits.** For a prime p let v be the Gauss valuation on $\mathbb{Q}[x]$ (the minimum of the p-adic valuations of the coefficients); it is multiplicative. Expanding $\det \Phi(x)$ by Leibniz, $-v(\det \Phi)$ is at most the value τ_p of the *assignment problem* with weights $W_{ab} = \max(0, -v_p(M[a,b]))$, i.e. the maximum of $\sum_a W_{a\sigma(a)}$ over bijections $\sigma : R \to Q$. By LP duality, any *potentials* u_a, v_b with $W_{ab} \leq u_a + v_b$ certify $\tau_p \leq \sum u_a + \sum v_b$. We give explicit potentials for every prime power p^e (they depend only on p^e/r and on $p \bmod 4$ or $p \bmod 6$), and sum over primes with the prime number theorem in arithmetic progressions. This gives a "tropical" content $\Lambda_{\mathrm{trop}} = 3.74993$ for λ_-.

The tropical bound is not sharp. If (u, v) is an optimal dual solution, the scaled matrix $B = p^u \Phi p^v$ has entries in the valuation ring $\mathcal{O}$ of v, and $v(\det \Phi) = -\tau_p + v(\det B) \geq -\tau_p + \operatorname{corank}(B \bmod p)$. So every unit of corank of the reduction of B modulo p over $\mathbb{F}_p(x)$ saves one factor of p. We compute this corank in several ranges of primes, using the block structure of the reduction (residue classes modulo p), Schur complements, the identity $H_{p-1-n} \equiv H_n \pmod{p}$ for harmonic numbers, an antisymmetry of Cauchy-type sums, and the Frobenius–König theorem. The savings lower the content from 3.74993 to 3.70343. The uniform rescaled-core argument of Section 5 adds the missing interval savings without changing the determinant family, giving content 3.68274. Finally, Section 6 shows that at primes where the leading block is not clean the saving has a second level: the next Smith coefficient is a weighted Cauchy-type matrix of small rank, which adds further factors of p. This lowers the content to 3.67885.

2.4. **New idea 3: nonvanishing through a commuting Jacobi matrix.** Fix a prime ℓ in a suitable residue class with $\ell \nmid q$. After scaling by ℓ the rows and columns of nodes beyond a threshold $K \approx \ell/4$, the matrix $\Phi(x)$ reduces modulo ℓ to a *bordered Cauchy–Gram* matrix:

$$\ell^{2h} \det \Phi(x) \equiv \text{unit} \times \det \begin{pmatrix} yI_K + N_S & W_Y \\ W_X^\top & 0 \end{pmatrix} \pmod{\ell},$$
$$y = \text{affine function of } x,$$

where $N_S = \sum_{j=1}^{S} c_j c_j^\top$ is a sum of rank-one Gram terms of Cauchy vectors $c_j = (1/(a+j))_{0 \leq a < K}$ and W_X, W_Y consist of further Cauchy vectors. The two members of the pair differ by one Gram term. Their reductions cannot vanish simultaneously unless a *bordered Krylov determinant*

$$k = \det\big[W_Y \,\big|\, v,\ Av,\ \ldots,\ A^{K-m-1}v\big], \qquad A = \Pi N_S, \quad v = \Pi c_S,$$

vanishes modulo ℓ, where Π is the projection onto $\ker W_X^\top$ along the span of W_Y. We prove that, with the nodes shifted by a parameter θ, $k(\theta)$ is a constant prime to ℓ times a product of powers of $\theta + c$ with $1 \leq c < \ell$. The proof is algebraic and uses a structure familiar from orthogonal polynomials: there is an explicit *Jacobi (tridiagonal) matrix* $\tilde{T}$ that commutes with Π and with A and acts on the Cauchy vectors by a three-term recurrence (a bispectral situation in the sense of Grünbaum and Haine [18]). Its LU factorisation, reversed to UL, is (up to a shift and a diagonal conjugation) the Jacobi matrix of the next smaller instance: a discrete Darboux, or Christoffel, transformation [43, 19]. Krylov determinants of commuting pairs satisfy a cross-ratio identity, chains of Cauchy vectors give products of Cauchy determinants, and the remaining Gram determinants are evaluated with Cauchy–Binet, Hankel determinants of moments, Newton's divided differences and the Chu–Vandermonde identity (which also proves the Pfaff–Saalschütz-type identity needed). One auxiliary determinant has no product formula and is shown to be nonzero by positivity. An induction on

the size closes. We never need the exact exponents: it suffices to know that the determinant is an ℓ-adic unit times powers of $\theta + c$ with $1 \le c < \ell$.

Because the pair of members is available for every S in a range, the prime ℓ can be chosen *after* the size r, anywhere in an interval $[\frac{3}{2}r, 4r]$. This removes the cost of locating ℓ from the exponent (for λ_- and λ_+), and the final exponent is simply c/η.

2.5. **New idea 4: decay through an energy certificate.** Moving Hermite's contour to two rays in the left half-plane writes $\omega\,\mathsf{L}_F[R, Q]$ as a Gram matrix $\int v_a v_b\, dm$ of a complex measure m (finitely many signed atoms at poles of F plus an exponentially small density on the rays). Andréief's identity with two families of functions (rows and columns) gives

$$|\det \mathsf{L}_F[R, Q]| \le \frac{1}{n!} \int |\mathsf{C}_R(z)|\, |\mathsf{C}_Q(z)| \prod_{i=1}^{n} d|m|(z_i),$$

where $\mathsf{C}_R, \mathsf{C}_Q$ are Cauchy determinants. Taking logarithms turns the integrand into a discrete logarithmic energy of the free points z_i against the nodes in an external field. We bound this energy from below by comparing with a *certificate measure* σ (point masses with dyadic coordinates, placed where an optimal dual measure is supported) through a regularised kernel $-\frac{1}{2}\log(|z - w|^2 + \rho^2)$, which is conditionally positive definite on finite sums by an elementary argument (Frullani's integral and Gaussian sums). Everything left is a finite list of inequalities between explicit functions on boxes, which we verify in interval arithmetic inside Lean. For λ_- and λ_+ the contour rays have slope $\frac{5}{2}$ and $\rho = 1/4096$ (Section 9.4). The same argument bounds every principal minor along the common block, so the determinant stays small at every point within e^{-cr} of ξ.

2.6. **The deduction.** Write $\eta = D - \Lambda$. The three estimates are combined in §10; the result is

$$\mu(\xi) \le c\Big(\frac{1}{\eta} + g\, s\Big),$$

where c is the window constant of the decay estimate, $g = \lim r/\ell$ for the sizes at which nonvanishing is proved, and s is the relative length of the interval in which ℓ must be found. For λ_- and λ_+ the size can be chosen independently of ℓ, so the term gs disappears and $\mu \le 7.3546/0.22672 < 32.44$. For $L(2, \chi_{-3})$, Section 8 uses a quarter border and a characteristic-zero selection of one common-node deletion; no prime has to be chosen, so the term gs again disappears. Together with an analytic certificate and the saving at small primes this gives $\mu \le (376269/50000)/(1485123059/320000000 - \Lambda_{16}) < 14.21$, with $\Lambda_{16} = 339258406689121/82516315939200$.

constant	content Λ	decay D	window c	$\eta = D - \Lambda$	gs	exponent
$G - 5\pi^2/48$	3.67885	3.90557	7.3546	0.22672	0	32.4392
$G + 5\pi^2/48$	3.67885	3.90557	7.3546	0.22672	0	32.4392
$L(2, \chi_{-3})$	Λ_{16}	4.641010	7.52538	0.529600	0	14.209567
$L(2, \chi_{-3}) \pm \frac{2}{3}\zeta(2)$	3.2312582	3.87115	6.78788	0.6398917	0	10.6078
$\psi'(\frac{a}{5}) - \zeta(2)$	3.675334	3.90557	7.3546196	0.230235	0	31.9439

2.7. **Formal verification.** Every step, including the arithmetic of the dual certificates, the closed-form evaluations behind the nonvanishing, the interval-arithmetic decay certificates and the prime number theorem in progressions modulo 4 and 6, is formalised in Lean 4 with Mathlib [11, 45]. The formal proofs are built on a shared core (rectangular determinants, Gauss valuations and assignment bounds, bordered Krylov determinants, energy certificates, the measure deduction) with thin instances for the three functions in (2.1). The written proofs below are at the level of a research article; for long computations we point to the formal proof, which is the basis for the correctness of Theorem 1.1.

3. The construction

3.1. **The kernels.** For the three functions of (2.1) the kernel M of (2.2) is as follows; here $0 \le a < b$ are integers and $M[b, a] = M[a, b]$.

$$\begin{aligned}
\lambda_- : \quad M[a,b] &= -\frac{1}{b-a}\sum_{k=a+1}^{b}\frac{1}{16k(4k-1)},\\
M[a,a] &= -\sum_{k=1}^{a}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big);\\
\lambda_+ : \quad M[a,b] &= \frac{1}{b-a}\sum_{k=a+1}^{b}\frac{3}{16k(4k-3)},\\
M[a,a] &= -\sum_{k=1}^{a}\Big(\frac{1}{(4k-3)^2} - \frac{1}{16k^2}\Big);\\
\chi : \quad M[a,b] &= \frac{1}{6(b-a)}\sum_{k=a+1}^{b} d(k),\\
M[a,a] &= -\sum_{k=1}^{a}\delta(k),
\end{aligned}$$

with $d(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$ and $\delta(k) = \frac{1}{2(3k-2)^2} - \frac{1}{3(2k-1)^2} + \frac{1}{36k^2} = -\frac{1}{6}d'(k)$.

Lemma 3.1. *For each of the three functions, $\omega \llcorner_F [R, Q] = M[R, Q] + \xi E_{R,Q}$ for all finite sets $R, Q \subset \mathbb{Z}_{\ge 0}$, with ω and ξ as in (2.1).*

Proof. From $\psi(w+1) - \psi(w) = 1/w$ one gets $F(k) - F(k-1)$ as an explicit rational function of k; for instance $F_{\lambda_-}(k) - F_{\lambda_-}(k-1) = \frac{1}{k} - \frac{1}{k-1/4} =$

$-\frac{1}{k(4k-1)}$, and telescoping gives the off-diagonal entries. Similarly $\psi'(w+1)-\psi'(w)=-1/w^2$ gives $F'(a)-F'(0)$. Finally $\omega\,F'(0)=\xi$: for λ_-, $\frac{1}{16}(\psi'(1)-\psi'(\frac{3}{4}))=\frac{1}{16}(\frac{\pi^2}{6}-\pi^2+8G)=\lambda_-$; for λ_+ use $\psi'(\frac{1}{4})=\pi^2+8G$; for χ use $\psi'(\frac{1}{2})=\frac{\pi^2}{2}$ and $\psi'(\frac{1}{3})=\frac{2\pi^2}{3}+\frac{9}{2}L(2,\chi_{-3})$, which give $-\frac{1}{6}F'_\chi(0)=\sum_{k\ge1}\delta(k)=\frac{1}{4}L(2,\chi_{-3})$. □

Note that $2\lambda_-=G-5\pi^2/48$, $2\lambda_+=G+5\pi^2/48$ and $4\xi_\chi=L(2,\chi_{-3})$, and $\mu(a\xi+b)=\mu(\xi)$ for rational $a\neq0$, b. So Theorem 1.1 is a statement about the three numbers ξ.

3.2. **Geometries.**

Definition 3.2. Let $r,m\ge0$ and $e\in\{0,1\}$. The *geometry* $\mathcal{G}(r,m,e)$ consists of the common nodes $C=[0,r)\cup\{r+2m:\ e=1\}$, the row nodes $X'=\{r+2i:0\le i<m\}$ and the column nodes $Y'=\{r+2i+1:0\le i<m\}$. Rows are indexed by $R=C\sqcup X'$, columns by $Q=C\sqcup Y'$. The *member* $e=0$ is $\mathcal{G}_1(r,m)$, the member $e=1$ is $\mathcal{G}_2(r,m)$. The matrix of the geometry is

$$\Phi(x)=\Phi_{\mathcal{G}}(x)=\big(M[a,b]+x\,\mathbf{1}[a=b\in C]\big)_{a\in R,\ b\in Q},$$

a square matrix of size $|C|+m$, and $P_{\mathcal{G}}(x)=\det\Phi_{\mathcal{G}}(x)\in\mathbb{Q}[x]$.

By Lemma 3.1, $\Phi_{\mathcal{G}}(\xi+t)=\omega\,\mathsf{L}_F[R,Q]+t\,E_{R,Q}$ for real t. The geometry with $m=0$, $e=0$ is the square Löwner matrix of size r. We use *admissible* geometries, with $m\approx r/8$ for λ_-,λ_+, with $r\le8m\le r+8$; for $L(2,\chi_{-3})$ we use $m\approx r/4$, with $r\le4m\le r+3$ (Section 8). All $o(1)$ terms below are uniform over admissible geometries as $r\to\infty$.

Lemma 3.3 (integrality)**.** *$P_{\mathcal{G}}$ has degree at most $|C|$. If $d_{\mathcal{G}}=\operatorname{den}P_{\mathcal{G}}$ and p/q is a fraction in lowest terms with $P_{\mathcal{G}}(p/q)\neq0$, then $|d_{\mathcal{G}}P_{\mathcal{G}}(p/q)|\ge q^{-|C|}$.*

Proof. The variable occurs only in the $|C|$ diagonal entries indexed by C, so each Leibniz term has degree at most $|C|$. Then $q^{|C|}d_{\mathcal{G}}P_{\mathcal{G}}(p/q)$ is a nonzero integer. □

Remark 3.4. When the $m\times m$ block $M[X',Y']$ is invertible, the Schur complement gives $P_{\mathcal{G}}(x)=\det M[X',Y']\cdot\det(xI_C+M_{\mathrm{S}})$ with $M_{\mathrm{S}}=M[C,C]-M[C,Y']M[X',Y']^{-1}M[X',C]$. Thus the hybrid determinant is, up to a constant, the characteristic polynomial of a rational matrix of size $|C|$, as in the square case. We do not need this factorisation: Lemma 3.3 only uses the degree bound.

4. Arithmetic: the content

In this section we prove the following.

Theorem 4.1 (content)**.** *For admissible geometries of scale r,*

$$\log\operatorname{den}P_{\mathcal{G}}\le(\Lambda+o(1))\,r^2,$$

where $\Lambda = 3.67885$ *for* λ_- *and* λ_+, *and* $\Lambda = 339258406689121/82516315939200 = 4.111410\ldots$ *for* $L(2, \chi_{-3})$ *(quarter-border geometry, Section 8).*

We give the details for λ_- and indicate the changes for λ_+ and for $L(2, \chi_{-3})$ at the end.

4.1. **Gauss valuations and assignment problems.** Fix a prime p. Let $\mathcal{O} \subset \mathbb{Q}(x)$ be the localisation of $\mathbb{Z}_{(p)}[x]$ at the prime ideal (p); it is a discrete valuation ring with uniformiser p and residue field $\mathbb{F}_p(x)$, and its valuation v restricted to $\mathbb{Q}[x]$ is the *Gauss valuation* $v(f) = \min_k v_p(f_k)$, $f = \sum f_k x^k$. By Gauss's lemma $v(fg) = v(f) + v(g)$, and $v_p(\operatorname{den} f) = \max(0, -v(f))$.

Lemma 4.2 (assignment bound and LP duality)**.** *Let* B *be a square matrix over* $\mathbb{Q}[x]$ *with rows* I *and columns* J, $W_{ab} = \max(0, -v(B_{ab}))$, *and let* $\tau(W) = \max_\sigma \sum_{a \in I} W_{a\sigma(a)}$ *over bijections* $\sigma : I \to J$. *Then* $v(\det B) \geq -\tau(W)$. *If real numbers* u_a, v_b *satisfy* $W_{ab} \leq u_a + v_b$ *for all* a, b, *then* $\tau(W) \leq \sum_a u_a + \sum_b v_b$.

Proof. Each Leibniz term has valuation $\sum_a v(B_{a\sigma(a)}) \geq -\sum_a W_{a\sigma(a)}$, and v of a sum is at least the minimum. The second statement is weak LP duality for the assignment problem: $\sum_a W_{a\sigma(a)} \leq \sum_a (u_a + v_{\sigma(a)})$. □

For $\Phi(x)$ the weight of every entry is $W_{ab}(p) = \max(0, -v_p(M[a, b]))$, since $v(x) = 0$. We call τ_p the value of this assignment problem; thus $v_p(\operatorname{den} P_{\mathcal{G}}) \leq \tau_p$.

4.2. **The level majorant.** Let $p \geq 5$ and, for a prime power $q = p^e$, let $S_q = \{k \geq 1 : q \mid k(4k-1)\}$; it consists of the classes $k \equiv 0$ and $k \equiv k_q \pmod q$, where $k_q = (j_q q + 1)/4$ with $j_q \in \{1, 3\}$, $j_q q \equiv 3 \pmod 4$. Let $\beta_q(a) = \#(S_q \cap [1, a])$. The ultrametric inequality applied to the formulas for M gives

$$W_{ab}(p) \leq \sum_{e \geq 1} w^{(p^e)}(a, b),$$

$$w^{(q)}(a, b) = \mathbf{1}[\beta_q(a) \neq \beta_q(b)] + \mathbf{1}[q \mid a - b] \ (a \neq b),$$

$$w^{(q)}(c, c) = 2 \cdot \mathbf{1}[c \geq k_q],$$

and $w^{(q)} \in \{0, 1, 2\}$. Since the maximum of a sum is at most the sum of the maxima, $\tau_p \leq \sum_e \tau(w^{(p^e)})$. The primes 2 and 3 and all levels $e \geq 2$ contribute $o(r^2)$ in total (each $w^{(q)}$ has assignment value $O(r)$, there are $O(\log r)$ levels for each of $2, 3$, and $\sum_{p^e \leq Cr,\, e \geq 2} \log p = O(\sqrt{r})$).

4.3. **Explicit dual certificates.** Fix an odd prime power q, write $k = k_q$, $N = r + 2m$, and let

$$\delta_A(q) = \min\bigl(2r,\ 2(r-n) + 2\min(n, (r-q)_+)\bigr), \qquad n = \min(r, k),$$

the value of the square block $C \times C$. Let $U = [r-q, k)$ (empty if $r - q \geq k$) be the set of "low unpaired" indices; a zone node $z \in X' \cup Y'$ is *low* if $z - q \in U$, and L is the number of low zone nodes.

Proposition 4.3 (dual certificates)**.** *For $\mathcal{G}(r,m,0)$ with $8m \le r$, $\tau(w^{(q)}) \le \delta_A(q) + E(q)$ with*

range	$E(q)$	*potentials*
$q < 2m$	$2m - q\,\|m/q\|$	*by residue class mod q*
$2m \le q \le r$	$\min(2m+L,\ m+L+\lvert U\rvert)$	*two certificates*
$r < q < N$	$2m + (N-q)$	$\mathbf{1}[c \ge k]$ *on C,* $1 + \mathbf{1}[z \ge q]$ *on the zone*
$q \ge N$, $k \le r$	$2m$	$\mathbf{1}[c \ge k]$ *on C,* 1 *on the zone*
$q \ge N$, $k > r$	$(N-k)_+$	0 *on C,* $\mathbf{1}[z \ge k]$ *on the zone*

($\|\cdot\|$ is the distance to the nearest integer.)

Proof sketch. In each range one writes down potentials and checks $w \le u+v$ entrywise, using two facts: two distinct indices not congruent modulo q have $w \le 1$, and two zone nodes are never congruent once $q \ge 2m$. For example, for $q \ge N$ and $k \le r$, give the common nodes the potential $\mathbf{1}[c \ge k]$ and the zone nodes the potential 1: there are no congruences, so $w \le 1$ off the diagonal, and the diagonal cells (only at $c \in C$) are covered. For $q < 2m$ the potentials are $\frac{1}{2}, \frac{3}{2}, 1$ on rows and $\frac{3}{2}, \frac{1}{2}, 1$ on columns of a residue class ϱ, according as X' has more, fewer or as many nodes in ϱ as Y'; since the zone alternates between X' and Y', the imbalance is at most one per class and the number of unbalanced classes is $2q\|m/q\|$. The certificates for $2m \le q \le r$ start from the square-frame potential on C and raise the potential of low zone nodes (first certificate), or move half a unit between rows and columns in the residue classes of high zone nodes (second certificate). All cases are checked in the formal proof. □

Remark 4.4. The certificates are sharp: the exact assignment value computed by the Hungarian algorithm agrees with $\delta_A + E$ for all odd q and all tested (r,m). Sharpness is not used.

4.4. **The tropical content.** Put $\theta = q/r$, $\alpha = m/r$. Proposition 4.3 gives $\tau(w^{(q)}) \le r\, f_j(\theta;\alpha) + O(1)$ for an explicit continuous piecewise linear function f_j supported in $[0, 4(1+2\alpha)/j]$. Summing over primes in the two classes modulo 4 with the prime number theorem in arithmetic progressions, $\frac{1}{r}\sum_{p \equiv a\,(4)} f(p/r)\log p \to \frac{1}{2}\int_0^\infty f$ for bounded Riemann-integrable f of compact support (see [10, Ch. 20] for the prime number theorem in progressions), one finds

$$\sum_p \tau_p \log p \le (\Lambda_{\mathrm{trop}}(\alpha) + o(1))r^2,$$

$$\Lambda_{\mathrm{trop}}(\alpha) = \frac{313}{105} + \frac{554}{105}\alpha + \Big(\frac{118}{21} + 2\log 2\Big)\alpha^2$$

for $\alpha \le \frac{1}{8}$ (the term $2\alpha^2 \log 2$ comes from the sawtooth $q\|m/q\|$ in the range $q < 2m$). At $\alpha = \frac{1}{8}$, $\Lambda_{\mathrm{trop}} = 3.7499347\ldots$; for $\alpha = 0$ it is the square-frame value $313/105$.

4.5. **Rank deficits.** The assignment bound is sharp for the matrix of weights, but not for $\det \Phi$: cancellation in the Leibniz expansion makes $\operatorname{den} P$ smaller. We capture it by the following principle.

Lemma 4.5 (rank deficit)**.** *Let (u, v) be integer-valued potentials feasible for the weights of Φ at p, of value τ'. Then $B = \operatorname{diag}(p^u)\,\Phi\,\operatorname{diag}(p^v)$ has entries in $\mathcal{O}$, and*

$$v_p(\operatorname{den} P_{\mathcal{G}}) \le \tau' - \operatorname{corank}_{\mathbb{F}_{p(x)}}(B \bmod p).$$

Proof. $v(\det \Phi) = -\tau' + v(\det B)$, and over the discrete valuation ring $\mathcal{O}$ the Smith normal form shows $v(\det B) \ge \operatorname{corank}(B \bmod p)$. (The formal proof avoids the Smith form: if $B \bmod p$ has rank ρ, an explicit rank witness and a change of basis over $\mathcal{O}$ exhibit $|I| - \rho$ rows divisible by p.) □

Write δ_p for the corank obtained in this way with an optimal dual (so $v_p(\operatorname{den} P) \le \tau_p - \delta_p$). We use five families of primes. In all of them $p^2 > 4N$, so only the level $e = 1$ matters; the reduction $B \bmod p$ has a block structure governed by the singular indices $k \in S_p$ below N, which cut $[0, N)$ into blocks on which the leading coefficients (lead_{-1} of the partial sums of $1/(4k(4k-1))$ and lead_{-2} of the diagonal sums) are constant.

Theorem 4.6 (rank deficits for λ_-)**.** *Let $\mathcal{G} = \mathcal{G}(r, m, e)$ be admissible and r large. Then $v_p(\operatorname{den} P_{\mathcal{G}}) \le \tau_p - \delta_p + O(1)$, with δ_p as follows, for all primes p except $p \in \{139, 1009\}$:*

primes	δ_p	*mechanism*
$p \equiv 3\ (4)$, $\frac{5}{8}r < p < \frac{4}{5}r$; $p \equiv 1\ (4)$, $\frac{5}{7}r < p < r$	m	*integral Schur complement*
$r < p < r + m$	$r + m - p$	*harmonic reflection, rank count*
$p \equiv 3\ (4)$, $\frac{4}{5}(r+m) \le p < r$	$\frac{5p+1}{4} - r - m$	*Frobenius–König*
$p \equiv 1\ (4)$, $\frac{4}{7}(r+m) \le p < \frac{5}{7}r$	$\frac{7p+1}{4} - r - m$	*Frobenius–König*
$p \equiv 3\ (4)$, $\frac{5}{12}r < p < \frac{4}{9}r$; $p \equiv 1\ (4)$, $\frac{5}{11}r \le p < \frac{1}{2}r$	m	*block-diagonal leading form*

We sketch the mechanisms; the complete proofs are formalised.

The integral Schur complement. Write $\Phi = \left(\begin{smallmatrix} A & \Phi[C,Y'] \\ \Phi[X',C] & \Phi[X',Y'] \end{smallmatrix}\right)$ with $A = \Phi[C, C]$, so $\det \Phi = \det A \cdot \det T$ with the Schur complement $T = \Phi[X', Y'] - \Phi[X', C]A^{-1}\Phi[C, Y']$. At these primes the square block has $-v(\det A) = 2r$,

and the claim is that T has entries in $\mathcal{O}$, whereas the assignment bound allows $v(T_{zy}) = -1$. For a zone row z with partner $c_z = z - p \in C$ decompose $\Phi[z, C] = A[c_z, C] + \ell_z$, and similarly for columns; since $A[c_z, \cdot]A^{-1} = e_{c_z}^\top$,

$$T_{zy} = \Delta^2 M(z, y) - \ell_z A^{-1} \ell_y^\top,$$
$$\Delta^2 M(z, y) = M(z, y) - M(z, c_y) - M(c_z, y) + M(c_z, c_y),$$

and x cancels. The leading form of $p^2 A$ is block diagonal by residue class (blocks of size 1 and 2), with explicit nonzero determinants (here $p \neq 139$ is used: one block has determinant $3^2 \cdot 5 \cdot 139/6400$). Expanding $\ell_z A^{-1} \ell_y^\top$ modulo p^2, the only surviving terms are $p\beta\big[\frac{\sigma}{4(c_z - c_y)} + \frac{\sigma}{4(c_y - c_z)}\big] = 0$, an antisymmetry of Cauchy-type sums. For $p \equiv 1 \pmod 4$ the analogous computation uses a scalar Schur complement: modulo p it equals $s_0 I$ with $s_0 = x - \frac{1}{16} H^{(2)}_{K_p - 1}$, $K_p = (3p + 1)/4$, which follows from partial fractions and the reflections $H_{p-1-n} \equiv H_n$, $H^{(2)}_{p-1-n} \equiv -H^{(2)}_n \pmod p$ for harmonic numbers. Here it is essential that x is an indeterminate: s_0 is a unit of $\mathcal{O}$, although it may vanish at a specialisation $x \in \mathbb{Q}$.

Primes just above r. For $r < p < r + m$ the only singular indices below N are k_p and p. After eliminating an invertible block, the Schur complement on $[0, k_p)$ is $s_0 I - \frac{1}{16} E E^\top$ with $E = (1/(u + i))_{u < k_p,\, 1 \le i \le p - r}$, by the same harmonic reflection. Each zone node $z < p$ contributes one rank-one term indexed by $i = p - z$, so the correction has rank at most $p - r$ and the corank is at least $r + m - p$.

Primes with one singular zone index. For $p \equiv 3 \pmod 4$ with $\frac{4}{5}(r + m) \le p < r$ (and $p \equiv 1 \pmod 4$ in the corresponding range) there is one singular index inside the zone. After elimination the zone×zone block of $B \bmod p$ has a zero sub-block of size $L_X \times L_Y$ with $L_X + L_Y = L \ge m$ (the low zone nodes), so by the Frobenius–König theorem its rank is at most $2m - L$, and the corank is at least $L - m$.

The low band. For primes p with $2m < p < r$ that are *clean* (no singular index in the zone) we prove a general statement: if every residue-class block of the leading form of $p^2 A$ is invertible modulo p, then T is integral. The proof splits $p^2 A = B_0 + pE$ with B_0 block diagonal by residue class, inverts by a Neumann series, and shows that the order-p term vanishes: the translation $c \mapsto c + (y - z)$ identifies the classes of z and y (cleanness is used here), and what remains is $\sum_{i,j} g_i g_j (L_i - L_j) = 0$, an antisymmetric sum. The original proof for the two listed families used classes of size 2 or 3 and explicit rational block determinants, for example $-350972849/368640000$ (with prime numerator), which are nonzero modulo p except at $p = 139$ and $p = 1009$. The uniform argument in Section 5 proves the needed clean-case bound without these exceptional determinants.

4.6. **Summation.** Each family is a union of intervals $p \in (ar, br)$ in a fixed class, and δ_p is a linear function of p there. With the prime number theorem

in progressions, the total saving is

$$\Big(\frac{129}{4480}+\frac{59}{4480}+\frac{29}{6336}\Big)r^2 = (0.0287946 + 0.0131696 + 0.0045770)\, r^2$$

(the three terms come from the first row, the next three rows, and the last row of the table). Hence

$$\Lambda_{\lambda_-} \le \Lambda_{\mathrm{trop}}(\tfrac{1}{8}) - 0.0465412 = 3.7033935\ldots \le 3.70343.$$

(The formal proof uses the slightly coarser rational value 3.7034204, from a truncation of the sawtooth term.) This is the intermediate content constant before the additional savings of Section 5, which complete the improved constant in Theorem 4.1.

4.7. **The second member.** The geometry $\mathcal{G}_2$ has one extra common node $b = r + 2m$. Bordering $\Phi_{\mathcal{G}_1}$ by one row and one column changes each per-prime bound by $O(1)$, so the content constant is unchanged; but the rank deficits have to be re-proved for the bordered matrix, since a deficit is not automatically inherited under bordering. For the deficits proved by rank witnesses there is a general bordering lemma: a witness for $\Phi_{\mathcal{G}_1}$ extends, with the border row and column scaled by p^w, to a witness for $\Phi_{\mathcal{G}_2}$ at a cost of $2w = O(1)$. For the low-band deficit, which is proved through integrality of the Schur complement, the border row and column are rescaled by p^w so that the same core argument applies, with one unit of slack.

4.8. **The mirror constant and $L(2, \chi_{-3})$.** For λ_+ the singular indices are those k with $p \mid k(4k-3)$; the two residue classes modulo 4 exchange their roles. The tropical law is the same function Λ_{trop}, the deficits are the same with the classes swapped, and $\Lambda_{\lambda_+} = \Lambda_{\lambda_-}$. The formal proof uses one generic content theorem with the offset (1 or 3) as a parameter.

For $L(2, \chi_{-3})$ the singular indices are those k with $p \mid k(3k-2)(2k-1)$, and the relevant classes of primes are modulo 6. Two explicit duals suffice: the residue-class potentials of the first row of Proposition 4.3, and a certificate in which only row potentials are raised, on nodes that have a congruent partner of level 0. The levels (leading coefficients of the partial sums) enter only through which of them vanish, and this reduces to eight explicit identities. The resulting bound is piecewise linear in p/r; the profile and its integral for the quarter-border geometry are given in Section 8.

5. A uniform rescaled-core deficit

The determinant family and the clean-prime estimates are those of Section 4. A common rescaled-core argument supplies three additional, disjoint groups of prime windows. The gain comes from a first-order cancellation and an elementary assignment dual; no assertion about a Smith normal form or the optimality of the tropical bound is required. The argument applies to both offsets $o \in \{1, 3\}$.

Monic rescaling. For the finite certificate take $m = \lfloor r/8 \rfloor$; the same leading estimates hold for $m = r/8 + O(1)$, since all endpoint and border losses are bounded per prime. Put $N = r + 2m$ and write $C = [0, r)$ for the common core and $Z = [r, N)$ for the interleaved border zone. We use primes $p \geq 13$ satisfying $4(N + 1) < p^2$. In the normalized symmetric rational matrix R, whose omitted normalization factors are p-adic units, every entry has valuation at least -2. Replace the polynomial variable X by x/p^2 and set

$$A = R_{CC} + \frac{x}{p^2} I_r, \qquad Q = p^2 A = p^2 R_{CC} + x I_r.$$

The entries of Q are integral for the p-adic Gauss valuation. Its reduction is block diagonal according to residues of core nodes modulo p. Every reduced block has the form $M + xI$, so its determinant is monic and nonzero over $\mathbb{F}_p(x)$. Consequently

$$v_G(\det Q) = 0, \qquad v_G(\det A) = -2r.$$

This remains valid when the constant matrix M is singular. The single Gauss unit $\det Q$ and its adjugate suffice to express all inverse entries; separate denominators for individual residue blocks are unnecessary.

Let T be the Schur matrix left after eliminating C, and let $w_{zy} \geq 0$ be integers with $v_G(T_{zy}) \geq -w_{zy}$. If every assignment in the $m \times m$ zone matrix has total weight $\sum w_{z\sigma(z)}$ at most τ, then the determinant expansion gives

$$v_G(F(x/p^2)) \geq -2r - \tau.$$

Indeed, for $F(X) = \sum_i c_i X^i$ the left side is $\min_i(v_p(c_i) - 2i)$. Thus every coefficient satisfies $v_p(c_i) \geq -2r - \tau + 2i \geq -2r - \tau$. The substitution estimates coefficients of the original polynomial and introduces no new approximation family.

Translation and cancellation. The singular cuts are the integers in the two residue classes 0 and $(cp + o)/4$ modulo p, where $c = 1$ or 3 is determined by the prime class. For a zone node z, its core chain consists of

$$\{z - ip : i \geq 1,\ 0 \leq z - ip < r\}.$$

Between consecutive singular cuts and the core-exit cuts $r + ip$, translation by $y - z$ identifies the chains of z and y, their leading node data, the leading core–zone vectors, and their symmetric leading chain matrices. For the one-interval construction below, clipping the right endpoint at $r + p$ controls the core-exit cut. The formal diagonal xI is preserved by the same identification.

For completeness, write $Q = B + pE$, where B retains the full same-residue blocks of Q. Modulo p^2,

$$Q^{-1} = B^{-1} - pB^{-1}EB^{-1}.$$

For two distinct residues the order -2 Schur term vanishes. For two nodes in one translated class, let g be the common leading inverse vector and L

the leading node datum. Up to one common normalization, the three order -1 correction sums are

$$\frac{\gamma}{y-z}\sum_u g_u(L(z)-L(u)),$$
$$\frac{\gamma}{y-z}\sum_u g_u(L(u)-L(z)),$$
$$-\frac{\gamma}{y-z}\sum_{u,v} g_u g_v(L(u)-L(v)).$$

The first two cancel termwise, and the last vanishes by interchanging u, v. The direct order -1 entry also vanishes since $L(y) = L(z)$. This proves integrality within the selected class. Between distinct residues the Schur entry has denominator exponent at most 1, and between equal residues at most 2. No specialization at $x = 0$ or additional spectral hypothesis enters this calculation.

One interval and a 0/1/2 dual. Choose consecutive singular cut indices $A_4 < B_4$, with no permitted numerator strictly between them, and put

$$a = \max\left(r, \left\lfloor \frac{A_4 p + \epsilon_{A_4}}{4} \right\rfloor\right),$$
$$b = \min\left(N, r+p, \left\lfloor \frac{B_4 p + \epsilon_{B_4}}{4} \right\rfloor\right),$$
$$\epsilon_j = \begin{cases} 0 & j \equiv 0 \pmod 4, \\ o & \text{otherwise.} \end{cases}$$

In the high range $p > 2m$ the clipping at $r + p$ is redundant. All nodes of $[a, b)$ lie in one translated class and have distinct residues. In the low range $m < p < 2m$, use the smaller interval

$$J = [\max(a, N-p), b),$$

interpreting its length as zero when its left endpoint exceeds its right endpoint. Since $b \le r + p$, a node in J has no other zone node of the same residue. In the high range take $J = [a, b)$. Assign potential 0 to every row or column node in J, and potential 1 to every other zone node. Two nodes in J have an integral Schur entry; a Schur entry between a node of J and a node outside J has denominator exponent at most 1, and one between two nodes outside J has denominator exponent at most 2. Therefore these potentials are a feasible dual for the assignment problem of Lemma 4.2, and

$$\tau \le 2m - |J|.$$

This single interval replaces the counting of residue classes and parities in Section 4.

The second member adds one row and one column to the Schur matrix. Every new entry, including the rescaled diagonal entry, has denominator exponent at most 2. An assignment uses at most two of the new entries, so

the bound increases by at most 4. Thus both members satisfy

$$\operatorname{denexp}_p F_i \le 2r + 2m - |J| + 4 \qquad (i = 1, 2),$$

where $\operatorname{denexp}_p F$ is the largest power of p in the denominators of the coefficients of F. The constant 4 is kept in the deficit below.

The deficit. Let T_4 be the bound for the denominator exponent at p used in Section 4, after the rank deficits of Theorem 4.6. In the ranges of primes considered here we compare it with

$$B_0 = \begin{cases} 2r + 3m - p, & m < p < 2m, \\ 2r + m, & p > 2m; \end{cases}$$

the affine bounds of Section 4 already give $T_4 \le B_0 + O(1)$ there, so the tropical integral is unchanged. The bound just proved gives the nonnegative integer deficit

$$\delta_p = \begin{cases} (m + |J| - p - 4)_+, & m < p < 2m, \\ (|J| - m - 4)_+, & p > 2m. \end{cases}$$

We use δ_p only at primes where the rank deficits of Theorem 4.6 give no saving, so that the two savings are never added at the same prime; elsewhere the new deficit is set to zero.

The profile. Write $u = p/r$, $A = A_4/4$ and $B = B_4/4$. The deficit is bounded below by $rf(p/r) - 8$, where f is a function of u that is affine on each of finitely many intervals. On each interval the inequality to be checked is, in the range $m < p < 2m$,

$$a_0 = \max(1, Au), \qquad b_0 = \min(5/4, Bu, 1 + u),$$
$$f(u) \le \min(1/8 - u + b_0 - a_0,\ b_0 - 9/8),$$

and in the range $p > 2m$,

$$f(u) \le \min(5/4, Bu) - \max(1, Au) - 1/8.$$

For affine f each of these reduces to finitely many inequalities at the endpoints: nine in the first range and four in the second. There are 61 intervals: 37 in the range $m < p < 2m$, 20 for non-clean primes with $u < 3/5$, and 4 further intervals listed below. For each interval one also checks that A_4 and B_4 are consecutive singular cuts, the residue class of p, that $f \ge 0$, and that the intervals of one class of primes are disjoint.

Each interval $[s, t]$ is shrunk to $[s + \eta, t - \eta]$ with $\eta = 10^{-7}$. This leaves a strict margin at the boundaries of the windows of Theorem 4.6, and a finite check shows that for all large r the shrunk intervals avoid those windows; this is how the two savings are kept apart. Replacing m by $r/8$ and the cuts by $A_4 p/4$, $B_4 p/4$ moves the endpoints by bounded amounts, and the explicit rounding bounds give, uniformly,

$$\delta_p \ge rf(p/r) - 8.$$

Both offsets use the same profile: the class $c = 1$ corresponds to $p \equiv 4 - o \pmod 4$ and $c = 3$ to $p \equiv o \pmod 4$, so passing from λ_- to λ_+ exchanges the two classes of primes without changing their contribution.

Prime summation and the final constants. For an affine piece $f(u) = \alpha + \beta u$ on $[s, t]$, the prime number theorem in progressions gives the contribution

$$\frac{1}{2}\int_s^t f(u)\,du = \frac{1}{2}\left(\alpha(t-s) + \frac{\beta}{2}(t^2 - s^2)\right)$$

to the coefficient of r^2; the constant 8 contributes only $O(r)$. Before shrinking, the three groups of intervals contribute

primes	intervals	contribution
$m < p < 2m$	37	$\frac{14667950664797}{5028641371353600} \simeq 0.0029168814$
non-clean	20	$\frac{160166243}{12415603200} \simeq 0.0129003996$
upper	4	$\frac{109}{22400} \simeq 0.0048660714$

The four upper affine pieces are

c	u-interval	$f(u)$
1	$(4/5, 9/10)$	$9/8 - 5u/4$
1	$(3/5, 5/8)$	$2u - 9/8$
3	$(5/8, 9/14)$	$9/8 - 7u/4$
3	$(3/5, 5/8)$	$u/4 - 1/8.$

After shrinking, the total of the 61 pieces is exactly

$$\Delta_\eta = \frac{4420349242147094110541}{213717258282528000000000} = 0.0206831646525407\ldots,$$

and the shrinking costs less than 1.879×10^{-7}. The content constant of Section 4, in the rational form used in the formal proof, is

$$\begin{aligned}\Lambda_4 &= \frac{24729491161}{6665218560} - \frac{1}{448} - \frac{1}{576} - \frac{1}{352} \\ &= \frac{320572813}{86561280} = 3.7034204323226274\ldots,\end{aligned}$$

and the new content constant is at most

$$\Lambda_4 - \Delta_\eta = 3.6827372676700867\ldots < \frac{184137}{50000} = 3.68274.$$

This is the content constant 3.68274 used in the next section; it uses no saving at primes $p < m$. The argument is first-order: it uses only the first Smith coefficient. Section 6 adds the second level at non-clean primes.

6. A next-level deficit at non-clean primes

Section 5 proves that the Schur complement of the zone is integral inside one cell and bounds the remaining entries by an assignment dual. This is a first-order statement. At a prime for which a long cell is overfull but does not contain the whole zone, the next Smith coefficient carries additional factors of p. We prove this here. It lowers the content constant from 3.68274 to 3.67885.

6.1. **Setting and statement.** Keep the notation of Section 5: $C = [0, r)$, $N = r + 2m$, $Z = [r, N)$, an offset $o \in \{1, 3\}$ and a prime $p \geq 13$ with $p > 2m$ and $4(N + 1) < p^2$. The zone rows are $r + 2i$ and the zone columns are $r + 2j + 1$, for $0 \leq i, j < m$. The singular residues 0 and $h \equiv (cp + o)/4 \pmod{p}$ cut $\mathbb{Z}/p\mathbb{Z}$ into two arcs. Call the arc with $(p \pm o)/4$ residues *short* and the other *long*. A *cell* is a maximal interval of integers in Z that contains no cut. It is *long* if its residues lie in the long arc. Let J be a long cell and put

$$\sigma = |J| - m.$$

The cell is *overfull* if $\sigma > 0$. It is *clean* if it contains the whole zone, that is, if $|J| = 2m$. Let S be the set of residues in the short arc, and let b be the number of residues in S that are not residues of zone nodes.

Theorem 6.1 (next-level deficit)**.** *Let J be an overfull long cell that is not clean. Then both members F_1, F_2 of the pair satisfy*

$$\operatorname{denexp}_p F_i \leq 2r + m - \min\bigl(m,\ \sigma + (\sigma - b)_+\bigr) + O(1),$$

where $(t)_+ = \max(t, 0)$ and the implied constant is absolute.

The bound of Section 5 is $2r + m - \sigma + O(1)$. The new term is $(\sigma - b)_+$. The truncation at m reflects only that valuations of the determinant are integers bounded by the size of the zone. A clean cell gets no new saving.

6.2. **The weighted congruence.** Write the normalized symmetric matrix R of Section 5 with the factor $n = 16$ and set $Y = U/p^2$, so that U is the monic variable. We work with the Gauss valuation v_G of $\mathbb{Q}(U)$ for the prime p. Its residue field is $\mathbb{F}_p(U)$. Split the unscaled prefix sums of the pole data as $F = F_{\mathrm{pole}} + F_{\mathrm{reg}}$ by keeping the terms whose denominators are divisible by p in the first part, and put $L = pF_{\mathrm{pole}}$. Under $4N < p^2$ every pole denominator has at most one factor p, so L and the scaled squared prefix are integral.

For a residue c let $Q_c = UI + p^2 R_{\mathrm{pol}}$ be the matrix on the core chain $\{k \in C : k \equiv c\}$, where R_{pol} is the Loewner matrix of the pole part. Its entries are Gauss integral and its determinant is monic in U. So $\det Q_c$ is a Gauss unit and Q_c^{-1} is Gauss integral. This uses the formal variable U and needs no specialization.

Fix a zone row z and a zone column y in the cell J, with distinct residues. Then $y - z$ is a unit. Let $g = Q_z^{-1} b$, where $b = p^2 R_{\mathrm{pol}}(\mathrm{chain}, z)$, and put

$$A = L(z) - \sum_i g_i L_i, \qquad B = 1 - \sum_i g_i,$$

the sums running over the chain of z. For every residue c put $u_c = A\mathbf{1} - BL_c$ and

$$\gamma(c) = u_c^{\top} Q_c^{-1} u_c.$$

Let γ_J be the value of γ at the chain of the source z.

Proposition 6.2 (weighted congruence)**.** *Let T be the Schur complement of the zone after elimination of C. For zone nodes z, y in the cell J with distinct residues, T_{zy} is Gauss integral, and modulo p*

$$T_{zy} \equiv \sum_{c \not\equiv z,y} \frac{w(c)}{(c-z)(c-y)}, \qquad w(c) = \frac{\gamma_J - \gamma(c)}{n^2} + \varepsilon \frac{B^2}{n} \mathbf{1}_S(c),$$

where ε is a unit constant depending only on the offset.

Sketch. Write $M_c = R_{CC} + (U/p^2)I$ for the core block and $r_z = R(z, C)$, $c_y = R(C, y)$. With G_z, G_y the extensions of g by zero off the chains of z and y, the Schur entry is exactly

$$T_{zy} = \Delta_{zy} - (r_z - G_z^{\top} M_c) M_c^{-1} (c_y - M_c G_y),$$

where $\Delta_{zy} = R_{zy} - G_z^{\top} c_y - r_z G_y + G_z^{\top} M_c G_y$. The chains of z and y are transported to each other by the translation $y - z$, since no cut lies between them. The exact commutator identities $n(t_i - t_j)(Q_z)_{ij} = L_j - L_i$ and $n t_i b_i = L(z) - L_i$, with $t_i = (\mathrm{node}_i - z)/p$, show that $Q_z(ntg) = u$. Therefore $\gamma = n(BJ - AI)$ with $I = -\sum t_i g_i$ and $J = -\sum t_i L_i g_i$. Reducing the cross terms chain by chain, with the block structure of M_c by residue, gives the displayed sum. The part with $\gamma_J - \gamma(c)$ comes from the pole prefix. The part with $\mathbf{1}_S$ comes from the regular prefix, which is periodic modulo p and equals an explicit cyclic sum on the short arc. □

6.3. **Long residues carry no weight.** Let c be a long residue. Then $w(c) = 0$ in each of the following cases.

(a) c is not a residue of a zone node. The chain of c has the same endpoint data as the chain of z, including the case that crosses r mod p, and the response is unchanged: $\gamma(c) = \gamma_J$.
(b) c is the residue of a zone node in the cell J. Chain transport inside the cell gives $\gamma(c) = \gamma_J$ again.
(c) c is the residue of a zone node outside J. Here $\gamma(c)$ may differ from γ_J. But we may project: the Cauchy kernels supported on J vanish at such a residue.

Hence, after projection, the sum in Proposition 6.2 runs over the b short residues not occupied by the zone. The remaining weights are arbitrary.

6.4. **From the congruence to the valuation.** Let W be the reduction modulo p of the leading coefficient of pT on the rows and columns of J. By Proposition 6.2, after the projection of the previous paragraph, W is a sum of at most b rank-one matrices $w(c)\,k_c(z)k_c(y)$, where $k_c(z) = 1/(c-z)$. So $\operatorname{rank} W \le b$.

The first-order argument of Section 5 shows that $\det(pT)$ has valuation at least σ. The two-level determinant lemma gives more. If σ units come from vanishing leading coefficients and the next coefficient matrix has rank at most b, then every unit of corank beyond b gives one more factor of p. Hence

$$v_G \det(pT) \ge \sigma + (\sigma - b)_+.$$

The lemma does not assume that the leading corank is exact, and it does not assume that the Cauchy coefficients are nonzero.

The second member. The matrix of F_2 has one more common node $q = N$ with the same formal diagonal variable. Since $0 < q - z \le 2m < p$, the residue of q differs from every zone residue. It changes exactly one chain response, $\gamma(q \bmod p)$. The reduction W changes by one outer product, so its rank grows by at most one. The new core node has determinant valuation -2. The total fixed loss is at most 3. The same bound for the leading block gives $v_G \ge \sigma + (\sigma - b - 1)_+$ for the second member. This proves Theorem 6.1.

6.5. **Summation.** Put $u = p/r$ and $m = \lfloor r/8 \rfloor$. The relevant primes have $u \in (\frac{1}{4}, \frac{3}{5})$, so that $p > 2m$. For a prime class $c \in \{1, 3\}$ and a given u, the new deficit is

$$f_{\mathrm{new}}(u) = \max_J \min\bigl(\tfrac{1}{8},\ \sigma_J + (\sigma_J - b_J)_+\bigr),$$

the maximum over overfull, non-clean long cells J at u, with lengths and b scaled by r. This function is piecewise affine. We merge it with the 61 rows of Section 5 by a pointwise maximum, so that no deficit is added twice at one prime. Each endpoint of each row is trimmed by 10^{-7}. Both offsets give the same profile with the two prime classes exchanged. The prime number theorem in progressions gives the quadratic contribution $\frac{1}{2}\int f\,du$ as in Section 5. The integral over the merged profile exceeds that of the 61 rows by at least 0.00389. Hence

$$\Lambda \le 3.68274 - 0.00389 \le 3.67885.$$

This is Theorem 4.1 for λ_- and λ_+.

Remark 6.3. The argument covers only overfull long cells, with z and y in one pole interval. Cells that contain the whole zone are excluded, and we claim no saving there.

7. Nonvanishing modulo a prime

This section follows the formal proof. We treat λ_- in detail; λ_+ and $L(2, \chi_{-3})$ are in §7.7. Throughout, $\ell \equiv 3 \pmod 4$ is a prime, $K = (\ell+1)/4$,

and $\mathcal{G}_e = \mathcal{G}(r, m, e)$, $e \in \{0, 1\}$, is the pair of members of a scale r, with

$$1 \le m < K \le r, \qquad S := \ell - r - 2m \;\ge\; n := K - m. \tag{7.1}$$

For fixed r and $r \le 8m \le r + 8$, (7.1) holds for every prime ℓ with $\frac{4}{3}(r + m) + \frac{1}{3} \le \ell \le 4r - 1$, an interval $[\frac{3}{2}r + O(1), 4r]$.

Theorem 7.1 (nonvanishing)**.** *Assume* (7.1) *and let* $x_0 \in \mathbb{Q}$ *with* $\ell \nmid \operatorname{den} x_0$. *Then* $P_{\mathcal{G}_0}(x_0) \neq 0$ *or* $P_{\mathcal{G}_1}(x_0) \neq 0$.

The proof has four parts: a reduction modulo ℓ to a pair of bordered Cauchy–Gram determinants (§7.1); a lemma showing that if both vanish, a bordered Krylov determinant vanishes (§7.2); a deformation in a parameter θ (§7.3); and an algebraic evaluation of the deformed Krylov determinant, up to factors that are units at ℓ, by a commuting Jacobi matrix, diagonal steps and Darboux steps (§§7.4–7.6).

7.1. **Reduction to bordered Cauchy–Gram determinants.** Call a node b *low* if $b < K$ and *high* if $K \le b \le \ell - 1$; all nodes of $\mathcal{G}_e$ are at most $r + 2m \le \ell - 1$, and all low nodes are common nodes since $K \le r$. For a high node put $j_b = \ell - b \in [1, \ell - K]$, and for $j \ge 1$ let $w_j = (1/(a + j))_{0 \le a < K} \in \mathbb{F}_\ell^K$; all denominators are nonzero modulo ℓ. For a set J let W_J be the $K \times |J|$ matrix with columns w_j, $j \in J$. In these coordinates the row nodes X' and the column nodes Y' have $J_{X'} = S + \{2, 4, \dots, 2m\}$ and $J_{Y'} = S + \{1, 3, \dots, 2m - 1\}$. The high nodes *absent* from the geometry are $\{1, \dots, S\}$ for $\mathcal{G}_0$ and $\{1, \dots, S - 1\}$ for $\mathcal{G}_1$, whose extra node $r + 2m$ has $j = S$. Put

$$N = \sum_{j=1}^{S-1} w_j w_j^{\top}, \qquad w = w_S.$$

Lemma 7.2 (ℓ-adic structure)**.** *The only ℓ-singular factor occurring in the entries $M[a, b]$, $a, b \le \ell - 1$, is $4K - 1 = \ell$. Consequently $M[a, b]$ is ℓ-integral if a, b are both low, or both high and distinct; for a low and b high, $\ell M[a, b] \equiv \frac{1}{4} w_{j_b}(a)$; for b high, $\ell^2 M[b, b] \equiv 1 \pmod{\ell}$; and on the low block,* $16\, M[\mathit{low}, \mathit{low}] \equiv W_{[1,\ell-K]} W_{[1,\ell-K]}^{\top} + H_{K-1}^{(2)} I_K \pmod{\ell}$, *where* $H_n^{(2)} = \sum_{i \le n} i^{-2}$.

Proof. $4k - 1 \in \{\ell, 2\ell, 3\ell\}$ with $k \le \ell - 1$ forces $k = K$. For a low and b high, the sum defining $M[a, b]$ contains the term $k = K$ once, and $16K = 4\ell + 4$, $b - a \equiv -(a + j_b)$. For the low block, $\frac{4}{4k-1} \equiv \frac{1}{k-K}$, and partial fractions give $\sum_{j=1}^{\ell-K} \frac{1}{(a+j)(b+j)}$ off the diagonal; on the diagonal one uses $H_{\ell-K}^{(2)} \equiv -H_{K-1}^{(2)}$. □

Proposition 7.3 (reduction)**.** *Let D be diagonal with entry 1 on low and ℓ on high rows and columns, and let h_e be the number of high rows of $\Phi_{\mathcal{G}_e}$.*

Then $D\Phi_{\mathcal{G}_e}(x)D$ is ℓ-integral, and with $y = 16x + H^{(2)}_{K-1}$,

$$\ell^{2h_1} P_{\mathcal{G}_1}(x) \equiv \pm 16^{-K} \det \mathcal{B}(y, N),$$
$$\ell^{2h_0} P_{\mathcal{G}_0}(x) \equiv \pm 16^{-K} \det \mathcal{B}(y, N + ww^\top) \pmod{\ell},$$

where $\mathcal{B}(y, N') = \begin{pmatrix} yI_K + N' & W_{Y'} \\ W_{X'}^\top & 0 \end{pmatrix}$.

Proof. By Lemma 7.2, modulo ℓ the matrix $D\Phi D$ has the block form $\begin{pmatrix} xI + M_{\mathrm{low}} & \frac{1}{4}W_H & \frac{1}{4}W_{Y'} \\ \frac{1}{4}W_H^\top & I & 0 \\ \frac{1}{4}W_{X'}^\top & 0 & 0 \end{pmatrix}$, where rows are low, high common, X', columns are low, high common, Y', and H is the set of j-coordinates of the high common nodes. Take the Schur complement of the identity block. By the low-block formula and the partition of $[1, \ell - K]$ into H, the absent set, $J_{X'}$ and $J_{Y'}$, the upper-left block becomes $\frac{1}{16}(yI + N_e + W_{X'}W_{X'}^\top + W_{Y'}W_{Y'}^\top)$, with N_e the Gram matrix of the absent columns. Row and column operations using the zero lower-right block remove the last two terms, and scaling gives the claim. □

Reduction modulo ℓ commutes with evaluation at $x_0 \in \mathbb{Z}_{(\ell)}$, and $x \mapsto y$ is the same for both members. So if $P_{\mathcal{G}_0}(x_0) = P_{\mathcal{G}_1}(x_0) = 0$, then for $y_0 = 16x_0 + H^{(2)}_{K-1} \in \mathbb{F}_\ell$ both $\det \mathcal{B}(y_0, N)$ and $\det \mathcal{B}(y_0, N + ww^\top)$ vanish.

7.2. **A pair of singular bordered matrices.** Let $\Delta = \det(W_{X'}^\top W_{Y'})$. If $\Delta \neq 0$, let $\Pi = I - W_{Y'}(W_{X'}^\top W_{Y'})^{-1}W_{X'}^\top$ be the projection onto $V = \ker W_{X'}^\top$ along the column span of $W_{Y'}$, and let U be a $K \times n$ matrix whose columns span V. Define

$$k_R = \det\big[\, w^\top (\Pi N)^k U \,\big]_{0 \le k < n},$$
$$k_L = \text{the same with } X' \text{ and } Y' \text{ exchanged.}$$

(Changing U multiplies k_R by a nonzero factor; in k_L the columns of U span $\ker W_{Y'}^\top$ instead.)

Lemma 7.4 (singular pair)**.** *Let N' be symmetric, $\tau \neq 0$, and suppose $\Delta \neq 0$. If $\det \mathcal{B}(y_0, N') = \det \mathcal{B}(y_0, N' + \tau ww^\top) = 0$ for some y_0, then $k_R = 0$ for every admissible U, or $k_L = 0$ for every admissible U (computed with N' in place of N).*

Proof. Put $\mathcal{B}_2 = \mathcal{B}(y_0, N')$ and $\hat{w} = (w, 0)$, and consider the bordered matrix $\mathcal{P} = \begin{pmatrix} \mathcal{B}_2 & \hat{w} \\ -\tau\hat{w}^\top & 1 \end{pmatrix}$. Its Schur complement on the corner 1 is $\mathcal{B}_2 + \tau\hat{w}\hat{w}^\top = \mathcal{B}(y_0, N' + \tau ww^\top)$, so $\det \mathcal{P} = 0$, and its principal minor $\mathcal{B}_2$ is singular. Then $\mathcal{B}_2$ has a right kernel vector $\hat{x}$ with $\hat{w}^\top\hat{x} = 0$, or a left kernel vector $\hat{x}$ with $\hat{x}^\top\hat{w} = 0$. (If $\operatorname{rank} \mathcal{B}_2$ is at most its size minus two, its kernel meets $\hat{w}^\perp$. Otherwise $\operatorname{adj} \mathcal{B}_2 = c\,\hat{x}\hat{y}^\top$ with $c \neq 0$ and $\hat{x}$, $\hat{y}$ spanning the right and left kernels, and $0 = \det \mathcal{P} = \det \mathcal{B}_2 + \tau\,\hat{w}^\top \operatorname{adj}(\mathcal{B}_2)\hat{w} = \tau c(\hat{w}^\top\hat{x})(\hat{y}^\top\hat{w})$.)

Take the right case, $\hat{x} = (x, c)$. Then $(y_0 + N')x + W_{Y'}c = 0$, $W_{X'}^\top x = 0$ and $w^\top x = 0$; and $x \neq 0$, since otherwise $W_{Y'}c = 0$ and $c = 0$ because $\Delta \neq 0$. Applying Π (which kills $W_{Y'}$ and fixes $x \in V$) gives $\Pi N' x = -y_0 x$, hence $w^\top (\Pi N')^k x = (-y_0)^k w^\top x = 0$ for all k. Writing $x = Uz$ with $z \neq 0$, the matrix $[w^\top (\Pi N')^k U]_k$ annihilates z, so $k_R = 0$. The left case is the same argument for the transpose, $\mathcal{B}(y_0, N')^\top = \mathcal{B}(y_0, N')$ with X' and Y' exchanged, and gives $k_L = 0$. □

No condition on y_0 is needed; in particular the value $y_0 = 0$ is not special. By Proposition 7.3 and Lemma 7.4 (with $N' = N$, $\tau = 1$), Theorem 7.1 follows from the three statements

$$\Delta \neq 0, \qquad k_R \neq 0, \qquad k_L \neq 0 \qquad \text{in } \mathbb{F}_\ell. \tag{7.2}$$

7.3. **Deformation and bordered Krylov determinants.** We lift to characteristic zero and deform the nodes. Let θ be a parameter, $c_y = (1/(\theta + a + y))_{0 \le a < K}$, W_X, W_Y the matrices with columns c_y for $y \in X = S + \{2, \dots, 2m\}$ and $y \in Y = S + \{1, \dots, 2m-1\}$, $N_s = \sum_{y=1}^{s} c_y c_y^\top$, Π the projection onto $V = \ker W_X^\top$ along $\operatorname{span} W_Y$, and $A_s = \Pi N_s$. For a $K \times K$ matrix B preserving V and $u \in V$, the *bordered Krylov determinant* is

$$\kappa(B, u) = \det\bigl[W_Y \,|\, u,\ Bu,\ \dots,\ B^{n-1}u\bigr];$$

it is nonzero exactly when u is a cyclic vector of $B|_V$. Put

$$k(\theta) = k_{K,m,S}(\theta) = \kappa(A_{S-1}, \Pi c_S) = \kappa(A_S, \Pi c_S)$$

(the two agree by (K2) below). At $\theta = 0$ all denominators are integers $a + y \le S + K + 2m - 1 = \ell - r + K - 1 < \ell$.

Lemma 7.5 (bridge)**.** *Let $K = m + n$ with $1 \le n \le S$, and let θ be generic, in the sense that $\theta + z \neq 0$ for every integer z, with $\Delta(\theta) \neq 0$. Then the row-Krylov determinant $\det[c_S^\top (\Pi N_{S-1})^k u_i]_{k,i<n}$, in which $u_1, \dots, u_n$ is the explicit basis U of V given by Lagrange interpolation at the nodes, equals $\nu(\theta)\, k(\theta)$, where $\nu(\theta) = C_\nu \prod_{a<K} \prod_{t \in Y} (\theta + a + t)$ and $C_\nu \in \mathbb{Q}^\times$ does not depend on θ. The constant C_ν is an ℓ-adic unit for every prime $\ell > S+K+m$ (all its prime factors are at most $S + K + m$).*

Proof sketch. Let U and V' be the Lagrange bases of $\ker W_X^\top$ and $\ker W_Y^\top$, $H = V'^\top U$ and $H_c = V'^\top N_{S-1} U$; both are symmetric. Then $\Pi = U H^{-1} V'^\top$, hence $\Pi N_{S-1} U = UB$ with $B = H^{-1} H_c$, and B is self-adjoint for H, which turns the row-Krylov matrix into the column-Krylov matrix up to the factors $(c_X/c_Y)^n$ and $\det H$, where $c_X = \Pi_X(S)/\prod_a(\theta + a + S)$ and c_Y is the same with Π_Y. Two block-determinant identities, $\det[W_X \,|\, V'] \det[W_Y \,|\, U] = \Delta \det H$ and $\det[W_Y \,|\, C] \det[W_X \,|\, V'] = \Delta \det(C^\top V')$ with $C = [c_1, \dots, c_n]$, eliminate $\det H$ and Δ. By Lagrange interpolation $C^\top V'$ is a diagonal matrix times a Vandermonde matrix, and $\det[W_Y \,|\, C]$ is a Cauchy determinant; this gives the stated ν, whose only θ-dependence is $\prod_{a<K} \prod_{t \in Y} (\theta + a + t)$. The

operator in the Krylov determinant of the formalisation is $\Pi N_S \Pi$, which agrees with A_S on V and is related to A_{S-1} by (K2). $\square$

Definition 7.6. Fix $B \geq 1$. A function f of θ is *good* (with bound B) if $f(\theta) = \varkappa \prod_{1 \leq c \leq B} (\theta + c)^{e_c}$ for rational $\theta > 0$, with integers e_c and a nonzero rational $\varkappa$ whose prime factors are at most B.

Good functions with bound B form a group under multiplication. The point of the definition is that a good function with $B < \ell$ is continuous at $\theta = 0$ and its value there is an ℓ-adic unit.

Theorem 7.7 (good form)**.** *Assume $1 \leq m < K$ and $S \geq n = K - m$. Then $k_{K,m,S}$ is good with bound $S + K + 2m - 1$.*

Theorem 7.7 is all that is needed. The exact exponents e_c are known (they are piecewise quadratic in c, and $\varkappa$ is a ratio of Hankel determinants; this was checked in exact arithmetic in 156 cases with $K \leq 9$), but the proof below never computes them.

Proof of Theorem 7.1 from Theorem 7.7 and Proposition 7.8. By Proposition 7.8 below, Δ is an ℓ-unit at $\theta = 0$, so $\Delta \neq 0$ in $\mathbb{F}_\ell$. The row-Krylov determinant of Lemma 7.5 is a continuous function of θ near 0 (its entries are rational in θ with no pole at 0, and Π is continuous because $\Delta(0) \neq 0$). By Lemma 7.5 and Theorem 7.7 it equals, for rational $\theta \in (0, 1)$, a good function with bound $S + K + 2m - 1 < \ell$, which is continuous at 0; hence the values at $\theta = 0$ agree, and the value is an ℓ-adic unit. All entries are ℓ-integral at $\theta = 0$, and reduction modulo ℓ of the matrices of §7.3 gives the matrices of §7.2 (after reversing the order of the border columns, which does not change Π). So $k_R \neq 0$ in $\mathbb{F}_\ell$, and the same argument with X and Y exchanged gives $k_L \neq 0$. This is (7.2). $\square$

Proposition 7.8 (the Cauchy-type Gram determinant)**.** *For $m \leq K$ and $\theta \notin \{-1, -2, \dots\}$,*

$$\det(W_X^{\top} W_Y) = V(X) V(Y) \, \frac{H_{K,m}}{\prod_{i<m} (\theta + S + 1 + i)_m \, (\theta + S + K + 1 + i)_m},$$

$$H_{K,m} = \det\Big[\sum_{a=0}^{K-1} a^{p+q} \Big]_{p,q<m},$$

where $V(X) = \prod_{s<s' \in X} (s' - s)$ and $V(Y)$ are Vandermonde products. The Hankel determinant $H_{K,m}$ of the uniform weight on $\{0, \dots, K-1\}$ is a product of factorials and binomial coefficients with all prime factors less than $2K$. In particular Δ is an ℓ-adic unit at $\theta = 0$ when $S + K + 2m \leq \ell$.

Proof. Step 1 (Cauchy–Binet). By Cauchy–Binet and Cauchy's determinant, $\det(W_X^{\top} W_Y)$ is $\pm V(X) V(Y)$ times $\sum_P V(P)^2 \prod_{a \in P} \rho(a)$ over m-subsets $P \subseteq \{0, \dots, K-1\}$, where $\rho(a) = \prod_{s \in X} (\theta + a + s)^{-1} \prod_{t \in Y} (\theta + a + t)^{-1}$. This sum is the Hankel determinant $\det[\sum_a \rho(a) a^{p+q}]_{p,q<m}$ of the moments of ρ.

Step 2 (basis change and telescoping). Replacing the monomials a^p and a^q by monic rising factorials $\varphi_p(a) = (a+\theta+1)_p$ and suitable ψ_q is a unitriangular change of basis, which does not change the determinant. After it, $\rho(a)\varphi_p(a)\psi_q(a) = 1/(a+\theta+p+1)_{2m-p-q}$, and these sums over $a<K$ telescope: $L\sum_{a<K} 1/(c+a)_{L+1} = 1/(c)_L - 1/(c+K)_L$. Hence each entry is a ratio of Gamma values times a divided difference $f[u_p, v_q]$ of $f(z) = (z)_K$ at two points $u_p = \theta+p+1$, $v_q = \theta+2m-q$.

Step 3 (Newton expansion). We use divided differences only through three facts: the recursion; Newton's expansion, which peels off the nodes $u_0, u_1, \dots$ one at a time and gives $f[u_p, v_q] = \sum_{i,j} N_i(u_p)N_j(v_q)\, f[\text{window}]$ with Newton basis polynomials N_i; and the values on windows of consecutive points, $f[s, s+1, \dots, s+n] = \binom{K}{n}(s+n)_{K-n}$ for $f = (z)_K$. Thus $[f[u_p, v_q]] = \mathcal{A}\,\mathcal{D}\,\mathcal{B}^\top$ with $\mathcal{A}$, $\mathcal{B}$ triangular, and the determinant reduces to a Pochhammer product times $\det[\binom{K}{p+q+1}]_{p,q<m}$.

Step 4 (Chu–Vandermonde). The identity $\sum_{a=0}^{K-1}\binom{a}{p}\binom{K-1-a}{q} = \binom{K}{p+q+1}$ (Chu–Vandermonde), together with another unitriangular basis change, gives $H_{K,m} = \prod_p (-1)^p p! \prod_q q!\, \det[\binom{K}{p+q+1}]$. No limit in θ is used. Finally $H_{K,m}$ is the Hankel determinant of the uniform weight, i.e. of the Hahn polynomials with $\alpha = \beta = 0$ [22, §9.5]; its classical product formula has all factors at most $K+m \le 2K-1$. □

7.4. **A commuting Jacobi matrix.** Put $g = S-n+1$ and, for $0 \le a < K$,

$$\sigma(a) = (a+1)(a+1+\theta)(K-1-a)\frac{\prod_{i=1}^{m}(\theta+S+2i+a)}{\prod_{i=1}^{m-1}(\theta+S+2i+1+a)},$$
$$\tau(a) = (a+1)(a+1+\theta)(K-1-a)\frac{\prod_{i=0}^{m}(\theta+S+2i+1+a)}{\prod_{i=1}^{m}(\theta+S+2i+a)},$$
$$\mu(a) = -(K-1-a)^2(\theta+S+1+a)(\theta+S+2m+1+a).$$

Let $\tilde{T}$ be the $K\times K$ tridiagonal matrix with $\tilde{T}_{a,a+1} = -\sigma(a)$, $\tilde{T}_{a+1,a} = -\tau(a)$ and

$$\tilde{T}_{aa} = \mu(a) + \frac{\sigma(a-1)\tau(a-1)}{\mu(a-1)} + g(g+1)a^2 + g\,c_0\,a,$$
$$c_0 = -2(K-1)(\theta+g) - 2K^2 + 2K - m^2.$$

For $m=0$ this is, up to conjugation, the Jacobi matrix of a family of Hahn polynomials; for $m \ge 1$ the products over i record the border.

Proposition 7.9 (bispectral identities)**.** *(i)* Column chain. *For* $1 \le y \le S$, $\tilde{T}c_y = l(y)\,c_{y-1} + d(y)\,c_y + r(y)\,c_{y+1} + \sum_{t\in Y}\kappa_t(y)\,c_t$ *with* $l(y) = -\tau(-\theta-y)$, $r(y) = -\sigma(-\theta-y-1)$, $d(y) = \tilde{T}_{aa}|_{a=-\theta-y}$, *and explicit corrections* $\kappa_t(y)$ *given by residues of* τ *and* σ *at the border points. The transpose* $\tilde{T}^\top$ *satisfies the analogous* row chain *with corrections in* $\operatorname{span} W_X$, *with the same coefficients.*

(ii) Invariance. $\tilde{T}$ *maps* $V = \ker W_X^\top$ *and* $\operatorname{span} W_Y$ *into themselves; hence* $\tilde{T}\Pi = \Pi\tilde{T}$.

(iii) Commutation. $[N_S, \tilde{T}]$ *maps* V *into* $\operatorname{span} W_Y$*; hence* $\tilde{T}$ *commutes with* A_S *on* V*.*

Proof sketch. (i) says that the three-term recurrence of $\tilde{T}$ in the node variable a is mirrored, under $a \mapsto -\theta - y$, by a three-term recurrence in the column variable y: a bispectral duality in the sense of Grünbaum and Haine [18]. It is an identity between rational functions of a. Conjugating by $\hat{Q}(z) = \prod_{t\in Y}(z+t)$ turns $\tilde{T}$ into a difference operator with polynomial coefficients, and (i) becomes the statement that a certain polynomial has degree at most $m+2$. Its four top coefficients are computed from the low-order coefficients of $\prod_i(1+c_iW)$ modulo W^4 and vanish identically; the remainder, of degree less than m, is identified by Lagrange interpolation at the border points. (ii) follows from (i) at the border points, where l and r vanish. (iii) follows from (i) and the row chain by telescoping $\sum_{y\le S}((\tilde{T}c_y)c_y^\top - c_y(\tilde{T}^\top c_y)^\top)$. □

7.5. **Krylov calculus.** We use five elementary facts about bordered Krylov determinants of matrices preserving V.

(K1) *Cross ratio.* If A and T commute on V, then $\kappa(A,u)\,\kappa(T,u') = \kappa(A,u')\,\kappa(T,u)$ for all $u, u' \in V$. (If $T|_V$ is cyclic, $A = p(T)$ on V, and $\kappa(A,u)/\kappa(T,u)$ is the determinant of the matrix of the powers of $p(T)$ in the Krylov basis of T, which does not depend on u.)

(K2) *Rank-one update along the start vector.* $\kappa(B + uz^\top, u) = \kappa(B,u)$.

(K3) *Chains.* If $Tc_y \equiv l(y)c_{y-1} + d(y)c_y + r(y)c_{y+1}$ modulo $\operatorname{span} W_Y$ for the relevant y, then $\kappa(T, \Pi c_{y_0}) = \prod_{k<n}\prod_{j<k} l(y_0 - j) \cdot \det[W_Y \mid c_{y_0}, c_{y_0-1}, \dots, c_{y_0-n+1}]$, and similarly upward with r; the Krylov vectors are triangular in the chain.

(K4) *Cauchy determinants.* $\det[W_Y \mid c_{y_1}, \dots, c_{y_n}]$ is a Cauchy determinant: a product of differences of the shifts divided by a product of factors $\theta + a + y$ [23, 24].

(K5) *Transfer.* If $B_2\Phi = \Phi B_1$ for a linear map Φ carrying V_1 into V_2, then $\kappa(B_2, \Phi z) = \det[W_2 \mid \Phi] \cdot \kappa(B_1, z)$, where $\det[W_2 \mid \Phi]$ is computed in adapted bases.

7.6. **Proof of Theorem 7.7.** Write $u(m,n,S;\theta) = k_{m+n,m,S}(\theta)$ and $B = S + n + 3m - 1 = S + K + 2m - 1$. We prove by induction on $n \ge 1$ that $u(m,n,S)$ is good with bound B for all $S \ge n$.

Base case $n = 1$. Then $\kappa(A, \Pi c_S) = \det[W_Y \mid \Pi c_S] = \det[W_Y \mid c_S]$, a Cauchy determinant, which is good.

Diagonal steps (from S *to* $S+1$*).* Since $c_y(\theta) = c_{y+1}(\theta - 1)$, the instances (θ, S) and $(\theta - 1, S+1)$ have the same border. Apply (K1) to the commuting pairs $(A, \tilde{T})$ of both instances, (K2) to pass between A_{S-1} and A_S, and (K3), (K4) to evaluate the four values $\kappa(\tilde{T}, \Pi c_y)$ that occur as explicit products (each is nonzero, so $\tilde{T}$ is cyclic). The result is

$$u(m,n,S+1;\theta-1) = u(m,n,S;\theta) \cdot \varkappa_{\mathrm{DS}} \prod_{j<n-1} \Big(\frac{\theta + K - 1 - j}{\theta + j}\Big)^{2(n-1-j)}$$

with a constant $\varkappa_{\mathrm{DS}}$ whose prime factors are at most B. Substituting $\theta+1$ for θ expresses $u(m,n,S+1;\theta)$ through $u(m,n,S;\theta+1)$ times factors $\theta+c$ with $1 \le c \le K$; so goodness passes from S to $S+1$. (It is this orientation, downward in S and upward in θ, that keeps all bases $c \ge 1$.)

Darboux steps (from $n-1$ to n at $S=n$). Consider the instance (K,m,S) with $S=n-1$, i.e. $g=0$. After normalising $\tilde T$ so that it annihilates the kernel vector of A on V, it has an LU factorisation

$$\tilde T = LU, \qquad U = \operatorname{diag}(\mu) - \operatorname{superdiag}(\sigma), \qquad L = I - \sum_a \frac{\tau(a-1)}{\mu(a-1)} E_{a,a-1},$$

with explicit pivots $\mu(a) = -(K-1-a)^2(\theta+S+1+a)(\theta+S+2m+1+a)$; note $\mu(K-1)=0$. Reversing the factors gives the Jacobi matrix of the instance with one node fewer and one more column:

$$(UL)_{<K-1} = D^{-1}(\tilde T' + \beta I)D, \qquad \tilde T' = \tilde T(K-1,m,S+1),$$

for an explicit diagonal D and scalar β. This is the discrete Darboux (Christoffel) transformation of Jacobi matrices [43, 49, 19]. The bidiagonal map $X = D\,U$ with its last row removed is an intertwiner:

$$X\tilde T = (\tilde T' + \beta)X, \qquad X\Pi = \Pi' X, \qquad XA = A'X,$$

where primes refer to $(K-1,m,S+1)$. The first identity is the reversal of the factors; the other two follow from the column and row chains of Proposition 7.9 at the border points ($Xc_t \in \operatorname{span} W_{Y'}$ for $t \in Y$, and dually) and, for A, from telescoping $XN_S - N'_{S+1}X$ in y. By (K5), (K2) and (K1),

$$\kappa(A,\Pi c_0)\,\kappa(\tilde T', w') = \det\bigl[W_Y \,|\, \Pi c_0 \,|\, E\bigr] \cdot \kappa(A',w') \cdot \prod_{k<n-1}\prod_{j<k} l(-j)\cdot \mathcal{G},$$

where $w' = \Pi' c'_n$ is a projected Cauchy vector of the smaller instance, E completes $[W_Y \,|\, \Pi c_0]$ by Cauchy columns (so the first factor is a Cauchy determinant), and $\mathcal{G} = \det[R^{\top}\Pi c_{-j}]_{j<n-1}$ with $R = [c_1,\dots,c_{n-1}]$. Two further inputs are needed.

(a) *The Gram determinant $\mathcal{G}$ (Chu–Vandermonde).* A Schur complement gives $\mathcal{G}\cdot\Delta = \det(C_{\mathcal{A}}^{\top} C_{\mathcal{B}})$ for two Cauchy matrices whose column sets $\mathcal{A}, \mathcal{B}$ have $K-1$ elements. By Cauchy–Binet this is a sum over the one omitted node a, with $V(P_a)^2 = V_K^2/(a!\,(K-1-a)!)^2$, and the sum is evaluated by

$$\sum_{a=0}^{N} \binom{N}{a}^2 (x+a)_{2N} = \binom{2N}{N}\bigl((x+N)_N\bigr)^2 \qquad (N = K-1).$$

This is a terminating balanced ${}_3F_2$ identity of Pfaff–Saalschütz type [1, §2.2]. We prove it with three applications of the Chu–Vandermonde identity: the moment identity $\sum_a \binom{N}{a}^2 a^{\underline{j}} = N^{\underline{j}}\binom{2N-j}{N}$ (falling factorials), the relation $\binom{2N}{i}\binom{2N-i}{N} = \binom{2N}{N}\binom{N}{i}$, and Vandermonde once more for $(x+2N-1)^{\underline{N}}$. With Proposition 7.8 this makes $\mathcal{G}$ explicit.

(b) *A nondegeneracy (positivity).* The transfer (K5) needs $\det[W' \,|\, X\Pi R] \neq 0$. Its numerator, as a function of θ, is not a product, so this is proved by positivity rather than by a formula: for rational $\theta > 0$ (this is the only range used, and it is proved over $\mathbb{Q}$), if $W's + X\Pi Rt = 0$, then $s = 0$ (apply $W_{X'}^{\top}$ and $X\Pi = \Pi' X$), so $y = \Pi Rt$ lies in $V \cap \ker X$; the range of $X^{\top}$ is contained in the span of $[R \,|\, W_X]$ (the row chain at the border points) and both have dimension $K-1$, so they are equal and $R^{\top}y = 0$, i.e. $(R^{\top}\Pi R)t = 0$. Finally $\det(R^{\top}\Pi R)\cdot\Delta = \det([R \,|\, W_X]^{\top}[R \,|\, W_Y])$ by a Schur complement. For two families of m' distinct shifts x, y with $a + x_i > 0$ and $a + y_i > 0$ for all nodes a, Cauchy–Binet gives

$$\det(C_x^{\top} C_y) = V(x)V(y) \sum_{P} V(P)^2 \prod_{a \in P} \prod_{i} \frac{1}{(a + x_i)(a + y_i)},$$

the sum over m'-subsets P of the nodes, with V the Vandermonde product. The sum is positive and $V(x)V(y) \neq 0$, so the determinant is nonzero. This applies to Δ and to $[R \,|\, W_X]^{\top}[R \,|\, W_Y]$, hence $\det(R^{\top}\Pi R) \neq 0$ and $t = 0$.

Every factor in the displayed identity is now good except the products over $l(-j) = -\tau(-\theta + j)$ and the Pochhammer denominator of $\mathcal{G}$, which contain factors $\theta - j - 1$ with negative shift; these two sets of factors coincide and cancel. The values $\kappa(\tilde{T}', w')$ and $\kappa(\tilde{T}, \cdot)$ are chain products by (K3) and nonzero. Combining with the diagonal step at $(K, m, n-1)$, (K1) and (K2), one obtains

$$u(m, n, n; \theta - 1) = u(m, n-1, n; \theta) \cdot (\text{good factor}).$$

Since $u(m, n-1, n)$ is good by the induction hypothesis (as $n \geq n-1$), so is $u(m, n, n)$, and the diagonal steps give all $S \geq n$. All bases stay in $[1, B]$ and all constants have prime factors at most B. This proves Theorem 7.7. □

7.7. **The mirror constant.** *The mirror constant.* For λ_+ one takes primes $\ell \equiv 1 \pmod 4$ and $K = (\ell + 3)/4$. The only ℓ-singular factor is again $4K - 3 = \ell$, and the reduction gives the same pair of bordered matrices (with the opposite sign of the shift). Everything else is unchanged.

8. A quarter border and characteristic-zero selection

For the conductor-three frame, characteristic-zero interpolation permits a larger border than a finite-field selection. We take

$$r \leq 4m \leq r + 3,$$

and retain the full determinant together with its principal restrictions obtained by omitting one common node. Their degree budget is still r. The denominator and decay estimates below apply uniformly to every member.

Theorem 8.1 (One common-node deletion)**.** *For every $r \geq 1$, $m \geq 0$ and rational input, the full conductor-three determinant or one of its principal*

common-node deletions is nonzero. No prime-band restriction on m/r is required.

Proof. Here is the interpolation argument underlying the certified selector. Write $N = r + 2m$, $n = r + m$, and use the r common nodes followed by the alternating row and column nodes in the consecutive tail. After the fixed normalization of the conductor-three frame, its divided differences have prefix values f_i, with

$$f_{i+1} - f_i = \delta(i), \qquad \delta(z) = \frac{1 - z}{216(z + 1)(z + 1/3)(z + 1/2)}.$$

The diagonal prefix e_i satisfies $e_{i+1} - e_i = -\delta'(i)$. Barycentric interpolation identifies either kernel with pairs (p, q) of degree less than n satisfying $q(i) = f_i p(i)$ at all N nodes, and $q'(c) = f_c p'(c) + (U - e_c)p(c)$ at the common nodes.

Two such pairs have zero cross product: its degree is at most $2n - 2$, while the value and first-jet conditions give $N + r = 2n$ zeros counted with multiplicity. If the kernel had dimension at least two, a nonzero pair could therefore be chosen with both degrees at most $n-2$. The discrete Wronskian

$$\begin{aligned} W(z) = 216(z + 1)(z + 1/3)(z + 1/2) \\ \cdot [q(z + 1)p(z) - q(z)p(z + 1)] \\ - (1 - z)p(z)p(z + 1). \end{aligned}$$

would have degree at most $2n - 3$, but its prescribed values and first jets supply $2n - 2$ zeros. Thus $W = 0$. This would make q/p a rational antidifference of δ, which is impossible: the simple pole at -1 is isolated in its integer-translation class. After cancellation of a numerator-denominator gcd, evaluating the cleared identity propagates a denominator root indefinitely in one direction, contradicting finite degree. The boundary case $n = 1$ is immediate.

If the full determinant is nonzero, retain it. Otherwise both kernels are one dimensional. A nonzero kernel pair cannot vanish at every common node. Dividing by their common node polynomial would give degree at most $m-1$ and the same value interpolation at all N nodes; its Wronskian would again vanish by degree and root count. The left and right barycentric maps land in the same one-dimensional interpolation-pair space. At a common node their coordinates are the same $p(c)$ multiplied by nonzero punctured-node factors. Thus the same common index is nonzero on both sides, and their adjugate outer product has a nonzero diagonal entry there. That principal cofactor is the required deletion. □

8.1. **The uniform arithmetic profile.** No new local valuation identity is needed: the two explicit dual certificates of Section 4 for $L(2, \chi_{-3})$ (the residue-class certificate and the certificate in which only row potentials are raised) suffice. At a prime p put $u = p/r$. We always use the first certificate, and the smaller of the two bounds when $r + 2m \le 2p$, the range in which the second is proved. For the two classes of primes modulo 6 this gives

the piecewise linear profiles w_1, w_5 below; at a common endpoint the two neighbouring pieces agree.

class	interval for u	$w(u)$
$1,5$	$[0,1/4]$	$5/2$
$1,5$	$[1/4,1/2]$	$11/4-u$
1	$[1/2,15/16]$	$9/4$
1	$[15/16,9/8]$	$7/2-4u/3$
1	$[9/8,3/2]$	$5-8u/3$
1	$[3/2,2]$	$2-2u/3$
1	$[2,3]$	$1-u/6$
1	$[3,9/2]$	$3/2-u/3$
5	$[1/2,5/6]$	$9/4$
5	$[5/6,1]$	$7/2-3u/2$
5	$[1,3/2]$	$4-2u$
5	$[3/2,2]$	$5/2-u$
5	$[2,3]$	$3/2-u/2$

Both profiles vanish beyond their last interval. Their integrals are $581/128$ and $355/96$, and the prime number theorem in the two classes gives

$$\Lambda_0=\frac{1}{2}\left(\frac{581}{128}+\frac{355}{96}\right)=\frac{3163}{768}.$$

For finite r the proof uses affine bounds with a fixed additive allowance for the rounding of m and of the residue endpoints; the contributions of prime powers and of the finitely many excluded primes are kept, and they are $o(r^2)$. The entrywise denominator clears every principal restriction, so this bound holds for the whole family with one deletion.

Small primes. For odd primes $p\le m$ the profile above ignores the bound

$$v_p(\mathrm{den})\le 2r+2m-\min(m \bmod p,\ p-m \bmod p),$$

which holds at every level of the nodes and comes from the distance of m to the nearest multiple of p. On each interval $m/(n+1)<p<m/n$, $n\ge 1$, it gives the additional saving $\min\{m-np,(n+1)p-m\}$. The two affine pieces meet at $p=2m/(2n+1)$, and

$$\int_{m/(n+1)}^{m/n}\min\{m-np,(n+1)p-m\}\,dp=\frac{m^2}{2n(n+1)(2n+1)}.$$

Using the intervals with $n\le 16$, the prime number theorem in the two classes modulo 6 gives the content constant

$$\Lambda_{16}=\Lambda_0-\frac{1}{16}\sum_{n=1}^{16}\frac{1}{2n(n+1)(2n+1)}=\frac{339258406689121}{82516315939200}.$$

The finite-size argument is the same for every n, with a fixed additive rounding allowance. The same denominator clears every principal restriction, so the family of determinants and the selection of Theorem 8.1 are unchanged.

8.2. **The analytic certificate.** The decay estimate of Section 9 is applied with $\alpha = 1/4$, the segment $[-\frac{107}{1000}, 0]$, contour rays of slope $\frac{12}{5}$, regularization radius $1/2048$, and a rational probability measure σ with 2950 atoms on the segment, the rays (in conjugate pairs) and the line, all of modulus at most 16. The positions and masses were found by solving the fractional program for $c/(D-\Lambda_{16})$ of §9.4 numerically, at the content of this section, and rounding to dyadic positions and masses. The function to be bounded is

$$g(s) = \frac{5}{4}U^{\sigma}(s) - U^{\nu_{1/4}}(s) + \Big(1 - \frac{1}{100000}\Big)\pi|\operatorname{Im} s|,$$

and the identity

$$U^{\nu_{1/4}}(s) = \frac{1}{2}U^{\tau}(s) + \frac{3}{4}U^{\tau}(2s/3) - \frac{3}{4}\log(3/2)$$

reduces the potential of the node distribution to the potential U^{τ} already bounded in Section 9. The inequalities on the rays, the segment and the line are checked on covers by cells, as in §9.4, with the near atoms of each cell treated one by one and the others by a Taylor expansion. With the sums over pairs of atoms this gives

$$S_{\text{base}} = -\frac{154489}{5000000}, \qquad I^{\sigma} \le \frac{179918953}{100000000}, \qquad I^{\nu} \ge \frac{184428}{100000}.$$

Hence $D \ge 1485123059/320000000 = 4.6410095\ldots$ and $c = 376269/50000 = 7.52538$. Omitting one common node changes the energy only by a term linear in r, which is absorbed in the $o(r^2)$ error. By Theorem 8.1 a retained determinant does not vanish at any rational point, so no prime has to be chosen, and the deduction of Section 10 gives

$$\mu(L(2,\chi_{-3})) \le \frac{376269/50000}{1485123059/320000000 - \Lambda_{16}} = 14.2095662593\ldots < 14.21.$$

9. Analytic decay

Theorem 9.1 (decay and window)**.** *For admissible geometries of scale r and real t with $|t| \le e^{-cr}$,*

$$|P_{\mathcal{G}}(\xi + t)| \le \exp(-(D - o(1))r^2),$$

$$(D, c) = \begin{cases} (3.90557,\ 7.3546) & \text{for } \lambda_-, \lambda_+, \\ (4.641010,\ 7.52538) & \text{for } L(2, \chi_{-3}). \end{cases}$$

We describe the proof for λ_-; the other two cases differ only in the reflection formula and the certificate data.

9.1. **A contour representation.** Hermite's formula gives $\mathsf{L}_F[a,b] = \frac{1}{2\pi i}\oint F(z)v_a(z)v_b(z)\,dz$ with $v_a(z) = 1/(z-a)$, the contour enclosing the nodes and no pole of F. The reflection formulas $\psi(1+z) = \psi(-z) - \pi\cot\pi z$ and $\psi(\frac{3}{4}+z) = \psi(\frac{1}{4}-z) + \pi\cot(\frac{\pi}{4} - \pi z)$ split $F = \tilde{F} + E$, where $\tilde{F}$ is analytic in the left half-plane and $O(1/|z|)$ there, and

$$E(z) = -\pi\big[\cot\pi z + \cot(\tfrac{\pi}{4} - \pi z)\big]$$

is 1-periodic with $|E(x+iy)| \le C_E e^{-2\pi|y|}$ for $|y| \ge 1$. Let $N_0 = \lfloor \frac{7}{50} r \rfloor$, $x^* = -N_0 - \frac{3}{8}$, and let s_0 be the slope parameter, $s_0 = \frac{5}{2}$ for λ_-, λ_+ (for $L(2, \chi_{-3})$ the slope is fixed by the certificate of Section 8), and let Γ be the two rays $x^* + u(-1 \pm s_0 i)$, $u \ge 0$. Moving the contour to Γ picks up the residues of F at the $2N_0$ poles $-(j + \frac{3}{4})$ (residue $+1$) and $-(j+1)$ (residue -1), $j < N_0$; the part of the integrand with $\tilde{F}$ integrates to 0 over Γ. Hence, for all integers $a, b \ge 0$,

$$\mathsf{L}_F[a,b] = \int v_a v_b \, dm, \qquad m = -\sum_{\mu} \epsilon_\mu \, \delta_{-\mu} + \frac{1}{2\pi i} E(z)\, dz\big|_\Gamma, \tag{9.1}$$

a Gram representation with respect to a complex measure m with finitely many atoms and an exponentially small density on Γ.

9.2. **Two-family Andréief bound.** For a set $S \subseteq C$ of common nodes let $R_S = S \sqcup X'$, $Q_S = S \sqcup Y'$, $n = |S| + m$, and $\mathcal{M}_S = \omega\, \mathsf{L}_F[R_S, Q_S]$. By multilinearity in the diagonal,

$$P_{\mathcal{G}}(\xi + t) = \sum_{S \subseteq C} t^{|C|-|S|} \det \mathcal{M}_S. \tag{9.2}$$

Andréief's identity [15] for the two families $(v_a)_{a \in R_S}$ and $(v_b)_{b \in Q_S}$ gives

$$\det \mathsf{L}_F[R_S, Q_S] = \frac{1}{n!} \int \det[v_a(z_i)]\, \det[v_b(z_i)] \prod_{i=1}^{n} dm(z_i),$$

and Cauchy's determinant evaluates both factors. Taking absolute values (see [42] for the logarithmic potential theory behind what follows),

$$|\det \mathsf{L}_F[R_S, Q_S]| \le \frac{1}{n!} \int \prod_{i<i'} |z_i - z_{i'}|^2 \prod_{\mathcal{F} \in \{R_S, Q_S\}} \frac{\prod_{a<a' \in \mathcal{F}} |a' - a|}{\prod_{i,\, a \in \mathcal{F}} |z_i - a|} \prod_i d|m|(z_i).$$

The integrand is the exponential of minus a discrete logarithmic energy: the free points z_i repel each other with weight 1, each node family repels itself with weight $\frac{1}{2}$ and attracts the free points with weight 1, and the density $|E| \approx e^{-2\pi |\operatorname{Im} z|}$ acts as an external field. Because the rows and columns differ only in the interleaved blocks, the two node families have the same macroscopic density: after scaling by r, both are

$$\nu_\alpha = \mathrm{Leb}|_{[0,1]} + \tfrac{1}{2}\mathrm{Leb}|_{[1,1+2\alpha]}, \qquad \alpha = \tfrac{1}{8},$$

of mass $1 + \alpha$. Its logarithmic energy has the closed form

$$I(\nu_{1/8}) = \iint -\log|u - u'| \, d\nu_{1/8}(u)\, d\nu_{1/8}(u') = \frac{243}{128} - \frac{\log 2}{32} - \frac{25}{32} \log \frac{5}{4} = 1.70244\ldots$$

9.3. **The energy certificate.** Fix ρ ($\rho = 1/4096$ for λ_-, λ_+ and $\rho = 1/2048$ for $L(2, \chi_{-3})$) and the regularised kernel $k_\rho(z, w) = -\frac{1}{2}\log(|z-w|^2+\rho^2) \le -\log|z-w|$.

Lemma 9.2 (conditional positive definiteness)**.** *For points $p_1, \dots, p_N \in \mathbb{C}$ and real weights q_α with $\sum q_\alpha = 0$, $\sum_{\alpha,\beta} q_\alpha q_\beta k_\rho(p_\alpha, p_\beta) \ge 0$.*

Proof. Frullani's integral $\log a = \int_0^\infty (e^{-t} - e^{-at})\,dt/t$ and $\sum q_\alpha = 0$ give $\sum q_\alpha q_\beta(-\log(|p_\alpha - p_\beta|^2+\rho^2)) = \int_0^\infty \frac{e^{-t\rho^2}}{t} \sum q_\alpha q_\beta e^{-t|p_\alpha - p_\beta|^2}\,dt$, and Gaussian kernels are positive definite: expanding $e^{2t\langle p, p'\rangle}$ in powers and the binomial theorem write the inner sum as a convergent sum of squares. □

Apply Lemma 9.2 to the free points and removed nodes (weight $1/r$ each) against a *certificate measure* $\sigma = \sum_k m_k \delta_{c_k}$ of total mass $1+\alpha$. This bounds the free–free energy from below by a sum of one-point terms $U^\sigma_\rho(s) = \sum_k m_k k_\rho(s, c_k)$ and the constant $I_\rho(\sigma) = \sum_{k,l} m_k m_l k_\rho(c_k, c_l)$. Riemann-sum comparisons replace the node sums by the potential U^{ν_α} and the node self-energy by $I(\nu_\alpha)$, up to $O(r\log r)$. With the external field one arrives at the function

$$g(s) = (1+\alpha)U^\sigma_\rho(s) - U^{\nu_\alpha}(s) + (1-\kappa_0)\pi|\operatorname{Im} s|, \qquad \kappa_0 = 10^{-3},$$

on the scaled support of $|m|$ (the rays Γ/r, the segment $[-\frac{7}{50}, 0]$ containing the atoms, and, for removed nodes, the segment $[0, 1+2\alpha]$). The certificate consists of three finite families of inequalities, checked in interval arithmetic:

(a) $g \ge m'$ on a cover of the rays by boxes, with an analytic tail bound beyond $u = \frac{3}{2}$;
(b) $g \ge \gamma_\ell$ on a partition of $[-\frac{7}{50}, 0]$ into cells; put $S_{\mathrm{cert}} = 2\sum_\ell |\mathrm{cell}_\ell| \min(\gamma_\ell - m', 0)$;
(c) $g \ge g_{\min}$ on a cover of $[0, 1+2\alpha]$.

On a box, U^σ_ρ is bounded below by moving each c_k to the farthest corner, and U^{ν_α} is bounded above from its closed form and monotonicity. Then

$$D = 2\big((1+\alpha)m' + S_{\mathrm{cert}}\big) - (1+\alpha)^2 I_\rho(\sigma) + I(\nu_\alpha), \qquad c = 2(m' - g_{\min}).$$

Proposition 9.3 (certified constants)**.** *For λ_- and λ_+ there is a measure σ with 1650 atoms with dyadic coordinates and masses, and $\rho = 1/4096$, for which*

$$m' \ge 2.1711, \quad S_{\mathrm{cert}} \ge -0.12573, \quad (1+\alpha)^2 I_\rho(\sigma) \le 2.43054,$$
$$g_{\min} \ge -1.5061, \quad I(\nu_{1/8}) \ge 1.7024457,$$

whence $D \ge 3.90557$ and $c \le 7.3546$. For $L(2, \chi_{-3})$ the certificate of Section 8 gives $D \ge 4.641009$ and $c \le 7.52538$.

All the inequalities are verified by the Lean kernel (§14); the data were generated with interval arithmetic in Arb [21] and then checked independently in Lean with rational arithmetic. The certificate for λ_- and λ_+ is described in the next subsection.

9.4. **Placement of the atoms and steeper rays.** Two choices distinguish the certificate for λ_- and λ_+.

The slope. Nothing in the contour argument requires slope 3. The bound $|E(x+iy)| \le C_E e^{-2\pi|y|}$ holds on any pair of rays $x^* + u(-1 \pm s_0 i)$ with $s_0 \ge 1$, and the Gram representation (9.1) is unchanged. The slope enters the certificate only through the box cover of the rays (a) and the tail bound beyond $u = \frac{3}{2}$. The vertical coordinate on a box is $s_0 u \in [s_0 u_1, s_0 u_2]$. With the other parameters fixed, a scan over $s_0 \in \{\frac{9}{4}, \frac{5}{2}, \frac{11}{4}\}$ gives the best ratio at $s_0 = \frac{5}{2}$. The decay estimate is proved with s_0 as a parameter. Its exponent formula does not depend on s_0; only the certified constants m', $g_{\min}$ and c do.

The atoms. The measure σ has 150 atoms on the segment $[-\frac{7}{50}, 0]$, 600 conjugate pairs on the rays and 300 atoms on the line $[0,1]$. The positions were not chosen on a uniform grid. The certified ratio is the value of a fractional program in the masses of σ. We solved it numerically by a cutting-plane method together with Dinkelbach's iteration. The dual of this program is a measure on the support of m, and its support shows where the constraints $g \ge m'$, $g \ge \gamma_\ell$ and $g \ge g_{\min}$ are active. We place atoms at the active points, at spacing about ρ, and re-solve for the masses. With $\rho = 1/4096$ this gives the constants of Proposition 9.3.

Each of the three families of inequalities is checked on a cover by cells with exact integer arithmetic (dyadic cell endpoints in units of 2^{-40}, outward rounded). In each group of consecutive cells the atoms close to the cell are treated one by one and the remaining ones by a first-order Taylor expansion. The covers have 29 701 ray boxes, 6000 segment cells and 6430 line cells. The lower bounds in Lean are 10^{-5} below the interval values m' and $g_{\min}$. This allowance lets far atoms be grouped and costs only $4 \cdot 10^{-5}$ in D.

Remark 9.4 (limits of this certificate). Weak duality bounds every certificate of this form from below. If μ is a probability measure on the support of m, then any admissible function g satisfies an inequality in the integrals $\int g\, d\mu$, and an exact rational dual μ proves a lower bound for the exponent c/η that holds for all σ. Such dual measures show how close a given certificate is to the best possible one of the same form; they are not part of the proof. Smaller ρ needs proportionally more atoms.

9.5. **Reflection formula for $L(2, \chi_{-3})$.** For F_χ the reflection formula $\psi(z+a) = \psi(1-a-z) - \pi\cot\pi(z+a)$ gives $F_\chi = \tilde{F}_\chi + E_\chi$ with

$$E_\chi(z) = \pi\Big[\tfrac{1}{3}\cot\pi(z+\tfrac{1}{3}) - \tfrac{1}{2}\cot\pi(z+\tfrac{1}{2}) + \tfrac{1}{6}\cot\pi(z+1)\Big],$$

which is 1-periodic and, because the weights $\frac{1}{3} - \frac{1}{2} + \frac{1}{6}$ sum to zero, decays like $e^{-2\pi|\operatorname{Im} z|}$. The contour argument and the Andréief bound are as above, with poles at $-(j+\frac{1}{3})$, $-(j+\frac{1}{2})$, $-(j+1)$. The certificate used for $L(2, \chi_{-3})$ is the one of Section 8, with $\alpha = \frac{1}{4}$ and $r \le 4m \le r+3$; the relation between r and m only changes the $O(r \log r)$ error terms.

9.6. **Node removal and the window.** The argument applies to every subset $S \subseteq C$: removing $j = \beta r$ common nodes turns them into points of the configuration that are not integrated over, and only common nodes can be removed, because a node removed from one family only would leave an uncancelled term of weight $\frac{1}{2}$. One obtains

$$|\det \mathcal{M}_S| \le \exp(-r^2(D - \beta c - o(1))) \qquad (\beta = |C \setminus S|/r),$$

uniformly in S. Inserting this into (9.2) with $|t|e^{cr} \le 1$ gives Theorem 9.1: the sum has at most $2^{|C|} = e^{O(r)}$ terms, each at most $e^{-(D-o(1))r^2}$. The second member is handled by viewing it as the pattern $(r, m+1)$ with one column node moved by one place, at an energy cost $O(r)$.

10. Proof of Theorem 1.1

10.1. $G \pm 5\pi^2/48$**.** Let $\xi = \lambda_-$ (the case λ_+ is identical, with primes $\ell \equiv 1 \pmod 4$). Let $\eta = D - \Lambda \ge 3.90557 - 3.67885 = 0.22672$, let $\nu > c/\eta$, and suppose $|\xi - p/q| < q^{-\nu}$ with q large; put $L = \log q$. Choose $\delta > 0$ with $\nu > c(1+\delta)/\eta$, let $r = \lceil (1+\delta)L/\eta \rceil$ and choose m with $r \le 8m \le r + 8$.

Choice of ℓ. By the prime number theorem in progressions, the primes $\ell \equiv 3 \pmod 4$ in $[\frac{4}{3}(r+m)+1, 4r-1]$ have product $\exp((\frac{5}{4} + o(1))r)$ (as $m \le \frac{r}{8} + 1$), which exceeds $q = e^L$ since $r \gg L$. So one of them does not divide q. By Theorem 7.1 with $x_0 = p/q$, there is $e \in \{0, 1\}$ with $P_{\mathcal{G}_e}(p/q) \ne 0$, where $\mathcal{G}_e = \mathcal{G}(r, m, e)$.

Lower bound. By Lemma 3.3, $|d_{\mathcal{G}_e} P_{\mathcal{G}_e}(p/q)| \ge q^{-(r+1)} = e^{-(r+1)L}$.

Upper bound. Write $p/q = \xi + t$ with $|t| < e^{-\nu L} \le e^{-cr}$ (as $cr \le c(1+\delta)L/\eta + c < \nu L$ for large q). By Theorems 4.1 and 9.1, $|d_{\mathcal{G}_e} P_{\mathcal{G}_e}(p/q)| \le e^{(\Lambda + o(1))r^2} e^{-(D-o(1))r^2} = e^{-(\eta - o(1))r^2}$.

Contradiction. Comparing, $(\eta - o(1))r^2 \le (r+1)L$; but $\eta r \ge (1+\delta)L$. Hence $\mu(\lambda_-) \le c/\eta$, and

$$\frac{c}{\eta} \le \frac{7.3546}{3.90557 - 3.67885} = 32.4392\ldots < 32.44.$$

Since $G - 5\pi^2/48 = 2\lambda_-$ and $G + 5\pi^2/48 = 2\lambda_+$, this gives the first two bounds of Theorem 1.1.

10.2. $L(2, \chi_{-3})$**.** Use Section 8 with at most one common-node deletion, and put

$$\eta = \frac{1485123059}{320000000} - \Lambda_{16} = 0.5295995572\ldots, \qquad c = \frac{376269}{50000}.$$

Let $\nu > c/\eta$, choose $\delta > 0$ so that $c(1+\delta)/\eta < \nu$, put $L = \log q$ and $r = \lceil (1+\delta)L/\eta \rceil$, and set $m = \lceil r/4 \rceil$. By Theorem 8.1, some retained member P_J (the full determinant or a principal restriction omitting one common node) has $P_J(p/q) \ne 0$. The tropical denominator $d_{r,m}$ clears its

coefficients, its degree is at most r, and the uniform quarter-border content and decay bounds give

$$q^{-r} \le |d_{r,m} P_J(p/q)| \le \exp(-(\eta - o(1))r^2)$$

whenever $|\xi_\chi - p/q| < q^{-\nu}$; the required window holds because $cr < \nu L$. This contradicts $\eta r \ge (1+\delta)L$. Consequently $\mu(\xi_\chi) \le c/\eta = 14.2095662593\ldots < 14.21$; rational scaling by four completes Theorem 1.1. □

Remark 10.1 (relation to linear independence criteria). The deduction is a one-dimensional version of the familiar scheme behind the criteria of Nesterenko [34] and their use by Ball and Rivoal [3]: the nonzero integer $q^{|C|} d_{\mathcal{G}} P_{\mathcal{G}}(p/q)$ plays the role of a small linear form with integer coefficients. The difference is that the lower bound for the form comes from an arithmetic nonvanishing statement (Theorem 7.1) rather than from an analytic asymptotic, and that the size r of the determinant is chosen in terms of q.

11. The numbers $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$

The functions of (2.1) are differences of two digamma functions whose arguments differ by a rational shift, and the shift decides which constant appears: the shifts $\frac14$ and $\frac34$ produce G together with a multiple of π^2, and the shifts $\frac13$ and $\frac23$ produce $L(2,\chi_{-3})$ together with a multiple of π^2. In this section we carry the method over to the second pair. Since $\pi^2 = 6\zeta(2)$, the numbers obtained are $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2) = L(2,\chi_{-3}) \pm \pi^2/9$, and we prove Theorem 1.2. The three estimates of Section 2 are needed again. The content is computed with the same dual certificates as in Section 4, now evaluated at the zeros of $k(3k-2)$ and $k(3k-1)$ modulo a prime power, which lie near $q/3$ and $2q/3$. Nonvanishing is obtained by the characteristic-zero argument of Section 8, and needs no condition on primes at all. The decay estimate of Section 9 is used with a certificate of the same form, adapted to the new content constant.

11.1. **Frames and kernels.** Let $s \in \{\frac13, \frac23\}$ and put

$$F_s(z) = \psi(z+1) - \psi(z+s), \qquad \omega = \tfrac19.$$

Write $L = L(2,\chi_{-3})$. From $\psi'(\frac13) = 9\sum_{k\ge0}(3k+1)^{-2}$, $\psi'(\frac23) = 9\sum_{k\ge0}(3k+2)^{-2}$ and $\sum_{n\ge1} n^{-2} = \pi^2/6$ one gets

$$\psi'(\tfrac13) = \frac{2\pi^2}{3} + \frac{9}{2}L, \qquad \psi'(\tfrac23) = \frac{2\pi^2}{3} - \frac{9}{2}L, \tag{11.1}$$

(the sum of the two is $\pi^2/\sin^2(\pi/3) = 4\pi^2/3$, and their difference is $9L$). As $\psi'(1) = \pi^2/6$,

$$\xi_{1/3} := \omega F'_{1/3}(0) = -\frac12\Big(L + \frac{\pi^2}{9}\Big), \qquad \xi_{2/3} := \omega F'_{2/3}(0) = \frac12\Big(L - \frac{\pi^2}{9}\Big). \tag{11.2}$$

Both are rational affine images of the numbers of Theorem 1.2, and $\mu(a\xi + b) = \mu(\xi)$ for rational $a \ne 0$ and b.

Put $e = 2$ for $s = \frac{1}{3}$ and $e = 1$ for $s = \frac{2}{3}$, so that $k - 1 + s = (3k - e)/3$. From $\psi(w + 1) - \psi(w) = 1/w$,

$$F_s(k) - F_s(k-1) = \frac{1}{k} - \frac{3}{3k-e} = -\frac{e}{k(3k-e)},$$
$$F_s'(k) - F_s'(k-1) = -\frac{1}{k^2} + \frac{9}{(3k-e)^2},$$

and telescoping gives, exactly as in Lemma 3.1, the identity (2.2) for all finite sets $R, Q \subset \mathbb{Z}_{\geq 0}$ with $\omega \mathsf{L}_{F_s}[R, Q] = M[R, Q] + \xi_s E_{R,Q}$ and the symmetric rational kernel

$$M[a,b] = -\frac{e}{9(b-a)} \sum_{k=a+1}^{b} \frac{1}{k(3k-e)} \quad (a < b),$$
$$M[a,a] = -\frac{1}{9} \sum_{k=1}^{a} \Big(\frac{1}{k^2} - \frac{9}{(3k-e)^2} \Big).$$

The geometries are those of Section 3: the common block $C = [0, r)$ and the interleaved borders X', Y' of m nodes with

$$r \leq 8m \leq r + 9, \tag{11.3}$$

that is $m = \frac{r}{8} + O(1)$, $\alpha = \frac{1}{8}$. For $J \subseteq C$ let $\Phi_J(x)$ be the matrix with rows $J \sqcup X'$ and columns $J \sqcup Y'$, entries $M[a, b]$, and x added on the diagonal cells $a = b \in J$; put $P_J(x) = \det \Phi_J(x)$. The members used are $J = C$ and the sets $J = C \setminus \{c\}$ with one common node removed, as in Section 8.

11.2. **Content.**

Theorem 11.1 (content)**.** *Let $s \in \{\frac{1}{3}, \frac{2}{3}\}$ and let r, m satisfy (11.3). For every $J \subseteq C$ with $|C \setminus J| \leq 1$ there is a positive integer $d_{r,m}$, independent of J, with $d_{r,m} P_J \in \mathbb{Z}[x]$ and*

$$\log d_{r,m} \leq (3.2312582 + o(1))\, r^2,$$

uniformly in m.

The entries of M have denominators dividing $k(3k - e)$ and $b - a$, so, for a prime $p \geq 5$, the valuation analysis of §4 applies with the polynomial $k(4k - 1)$ replaced by $\Phi(k) = k(3k - e)$. For a prime power $q = p^f$ the set $S_q = \{k \geq 1 : q \mid \Phi(k)\}$ consists of the classes $k \equiv 0$ and $k \equiv k_q \pmod q$, where $k_q \in (0, q)$ solves $3k_q = e + j_q q$ with $j_q \in \{1, 2\}$. Thus k_q/q is, up to $O(1/q)$, equal to $\frac{1}{3}$ or $\frac{2}{3}$ according to the class of q modulo 3: $j_q = 1$ if $q \equiv 1 \pmod 3$ and $e = 2$, or if $q \equiv 2 \pmod 3$ and $e = 1$, and $j_q = 2$ otherwise. The ultrametric inequality gives the level majorant $w^{(q)}$ of §4 (the argument there uses only that S_q consists of the classes 0 and k_q, and that at most one of k and $3k - e$ is divisible by p), and Proposition 4.3 bounds the assignment value $\tau(w^{(q)})$ by an explicit function of r, m, q and k_q only. We do not repeat the potentials. We checked the resulting bound against the Hungarian algorithm for $(r, m) \in \{(40, 5), (48, 6), (56, 7), (64, 8)\}$

and all primes $q \le 5(r+2m)$, with equality in every case; this check is not part of the formal proof.

Put $\theta = q/r$. For $\theta \le \frac{1}{4} = 2\alpha$ the bound is the same as for the shifts $\frac{1}{4}, \frac{3}{4}$, since it does not involve k_q; its integral is $\frac{1}{2} + \frac{1}{32} + \frac{\log 2}{32}$. For $\theta > \frac{1}{4}$ it is piecewise linear in θ, with the following pieces, which are the minima of the explicit potentials of Proposition 4.3 on each range:

	$k_q \approx q/3$	$k_q \approx 2q/3$
	$\theta \in (\frac{1}{4}, \frac{27}{32}]$: $\frac{17}{8}$	$\theta \in (\frac{1}{4}, \frac{27}{40}]$: $\frac{17}{8}$
	$(\frac{27}{32}, \frac{15}{16}]$: $\frac{13}{4} - \frac{4}{3}\theta$	$(\frac{27}{40}, \frac{3}{4}]$: $\frac{13}{4} - \frac{5}{3}\theta$
	$(\frac{15}{16}, 1]$: $\frac{9}{2} - \frac{8}{3}\theta$	$(\frac{3}{4}, 1]$: $\frac{9}{2} - \frac{10}{3}\theta$
	$(1, \frac{5}{4}]$: $\frac{7}{2} - \frac{5}{3}\theta$	$(1, \frac{5}{4}]$: $\frac{7}{2} - \frac{7}{3}\theta$
	$(\frac{5}{4}, 3]$: $\frac{9}{4} - \frac{2}{3}\theta$	$(\frac{5}{4}, \frac{3}{2}]$: $\frac{9}{4} - \frac{4}{3}\theta$
	$(3, \frac{15}{4}]$: $\frac{5}{4} - \frac{1}{3}\theta$	$(\frac{3}{2}, \frac{15}{8}]$: $\frac{5}{4} - \frac{2}{3}\theta$
integral over $\theta > \frac{1}{4}$	$\frac{1809}{512}$	$\frac{1167}{640}$

and the bound vanishes beyond the last piece. By the prime number theorem in arithmetic progressions modulo 3 (we use the classes modulo 6), primes with $k_q \approx q/3$ and with $k_q \approx 2q/3$ have density $\frac{1}{2}$ each, so that

$$\sum_p \tau_p \log p \le (\Lambda_{\mathrm{trop}} + o(1))r^2,$$

$$\Lambda_{\mathrm{trop}} = \frac{1}{2}\Big(\frac{1809}{512} + \frac{1167}{640}\Big) + \frac{1}{2} + \frac{1}{32} + \frac{\log 2}{32} = \frac{16433}{5120} + \frac{\log 2}{32} = 3.23123\ldots$$

The primes 2 and 3, the prime powers with $f \ge 2$ and the finitely many exceptional primes contribute $o(r^2)$, as in §4; the contributions of the removal of one common node and of the rounding of m are $O(r \log r)$. The two kernels give the same Λ_{trop} because their sets of classes, $\{\frac{1}{3}, \frac{2}{3}\}$ with density $\frac{1}{2}$ each, coincide. (Computed with exact rational arithmetic for several $\alpha \le \frac{1}{8}$, the same procedure gives $\Lambda_{\mathrm{trop}}(\alpha) = \frac{103}{40} + \frac{179}{40}\alpha + (\frac{77}{16} + 2\log 2)\alpha^2$, whose value at $\alpha = \frac{1}{8}$ is the number above; only the value at $\alpha = \frac{1}{8}$ is used.) In the formal proof the piecewise linear majorants are rational, with a small additive allowance for the rounding of m and of the residue classes; their total is exactly $\Lambda = \frac{43074084803}{13330437120} = 3.2312582\ldots$, and this is the value used below. This proves Theorem 11.1. We have not looked for rank deficits (§4) for these two numbers, so 3.2312582 is the tropical bound only.

11.3. **Nonvanishing.** For these frames nonvanishing is unconditional and is obtained by the characteristic-zero argument of Section 8, which we recall in the form needed.

Theorem 11.2. *Let $s \in \{\frac{1}{3}, \frac{2}{3}\}$. For every $r \ge 1$, $m \ge 0$ and $x \in \mathbb{Q}$ there is a set $I \subseteq C = [0, r)$ with $|I| \le 1$ such that $P_{C \setminus I}(x) \ne 0$.*

Proof. The proof of Theorem 8.1 uses the frame only through the prefix increments of the divided differences and of the diagonal. For F_s they are

$$f_{i+1} - f_i = \delta_s(i), \qquad e_{i+1} - e_i = -\delta_s'(i),$$
$$\delta_s(z) = \frac{1}{9}\Big(\frac{1}{z+1} - \frac{1}{z+s}\Big) = \frac{s-1}{9(z+1)(z+s)},$$

that is, $\delta_s = A/D$ with A constant and $D = (z+1)(z+s)$ quadratic, instead of the cubic denominator of the frame F_χ. The interpolation argument for the kernels of the full determinant, the count of zeros of the discrete Wronskian (of degree at most $2n-3$, against $2n-2$ prescribed zeros), and the selection of a common node at which both adjugate factors are nonzero, are therefore unchanged. The one fact on which the conclusion rests is that δ_s has no rational antidifference: the simple pole at $z = -1$ is isolated in its class modulo $\mathbb{Z}$, because the other pole $z = -s$ satisfies $-s \notin -1 + \mathbb{Z}$ for $0 < s < 1$. Hence the denominator cleared from an antidifference would have a root propagating indefinitely in one direction, contradicting finite degree. □

No prime ℓ enters, so there is no condition on a window of primes, in contrast with the nonvanishing of Section 7, and no condition on q beyond $x = p/q$ being rational.

11.4. **Decay.** The reflection formula $\psi(w) = \psi(1-w) - \pi\cot\pi w$ gives $F_s = \tilde{F}_s + E_s$ with

$$\tilde{F}_s(z) = \psi(-z) - \psi(1-s-z), \qquad E_s(z) = \pi\big[\cot\pi(z+s) - \cot\pi z\big].$$

Here $\tilde{F}_s$ is analytic in the left half-plane and $O(1/|z|)$ there, and E_s is 1-periodic with $|E_s(x+iy)| \le C_s e^{-2\pi|y|}$ for $|y| \ge 1$. The poles of F_s are at $-(j+1)$ with residue -1 and at $-(j+s)$ with residue $+1$, $j \ge 0$. This is the situation of §9, with the pole $-(j+\frac{3}{4})$ replaced by $-(j+s)$. Moving the contour of Hermite's formula to the rays $x^* + u(-1 \pm \frac{5}{2}i)$ with x^* at distance at least $\frac{1}{3}$ from every pole gives the Gram representation (9.1) with atoms at the $2N_0$ poles, $N_0 = \lfloor \frac{7}{50} r \rfloor$. The Andréief bound and the comparison with the logarithmic potential of $\nu_{1/8}$ involve the measure m only through the macroscopic density of its atoms (two unit masses per unit length on $[-\frac{7}{50}r, 0]$) and through the decay rate $e^{-2\pi|y|}$ of the density on the rays. The offset s changes the position of the atoms within unit cells, an $O(1)$ shift, and the constant C_s; both only affect the $o(1)$ terms. The certificate of §9.4 (slope $\frac{5}{2}$, $\rho = 1/4096$) is therefore valid unchanged, with $D = 3.90557$ and $c = 7.3546196$; these give the exponent 10.907.

A certificate trades D against c, and the best trade depends on Λ: the masses of σ are chosen to minimise $c/(D-\Lambda)$, and the content of these frames is smaller than that of $G \pm 5\pi^2/48$. We therefore re-solve the fractional program of §9.4 with the same $\alpha = \frac{1}{8}$, slope $\frac{5}{2}$, segment $[-\frac{7}{50}, 0]$ and $\rho = 1/4096$, and with objective $c/(D-\Lambda)$ at the value of Λ above. The new

measure σ has 250 atoms on the segment, 1200 conjugate pairs on the rays and 300 atoms on the line. Its support reaches $|\operatorname{Im} z| \approx 1.42$, beyond the rectangle used for $\lambda_\pm$, so the tail bound is proved on the rectangle $[-1,1] \times [-\frac{3}{2},\frac{3}{2}]$. The three families of inequalities are checked as in §9.4, on covers with 70 020 ray boxes, 24 576 segment cells and 16 119 line cells. We obtain the following.

Theorem 11.3 (decay and window)**.** *For $s \in \{\frac{1}{3},\frac{2}{3}\}$, admissible geometries (11.3), every $J \subseteq C$ with $|C \setminus J| \le 1$ and real t with $|t| \le e^{-cr}$,*

$$|P_J(\xi_s + t)| \le \exp\bigl(-(D - o(1))r^2\bigr),$$
$$D = 3.87115 = \tfrac{77423}{20000}, \quad c = 6.78788 = \tfrac{169697}{25000}.$$

The decay estimate depends on the offset s only through the $o(1)$ terms, so the frames enter the constants only through the choice of σ. Removing one common node costs $e^{O(r)}$, as in the node-removal paragraph of §9.

11.5. **Proof of Theorem 1.2.** Let $\xi = \xi_s$ and

$$\eta = D - \Lambda = 3.87115 - 3.2312582 = 0.6398917,$$
$$\frac{c}{\eta} = \frac{6.78788}{3.87115 - 3.2312582} = 10.6078\ldots < 10.61.$$

Let $\nu > c/\eta$ and suppose $|\xi - p/q| < q^{-\nu}$ with q large; put $L_q = \log q$, choose $\delta > 0$ with $c(1+\delta)/\eta < \nu$, set $r = \lceil (1+\delta)L_q/\eta \rceil$ and choose m with (11.3). By Theorem 11.2 with $x = p/q$ some member P_J with $|C \setminus J| \le 1$ satisfies $P_J(p/q) \neq 0$. Since $d_{r,m}P_J$ has integer coefficients and degree at most r, $|d_{r,m}P_J(p/q)| \ge q^{-r}$. On the other hand, with $t = p/q - \xi$ we have $|t| < e^{-\nu L_q} \le e^{-cr}$, and Theorems 11.1 and 11.3 give

$$q^{-r} \le |d_{r,m}P_J(p/q)| \le e^{(\Lambda + o(1))r^2} e^{-(D - o(1))r^2} = e^{-(\eta - o(1))r^2}.$$

Hence $(\eta - o(1))r^2 \le rL_q$, contradicting $\eta r \ge (1+\delta)L_q$. So $\mu(\xi_s) \le c/\eta < 10.61$. In detail, $\eta = 3.87115 - 3.2312582\ldots = 0.6398917\ldots$ and

$$\frac{6.78788}{0.6398917\ldots} = 10.60785\ldots < 10.61.$$

By (11.2), $L + \pi^2/9 = -2\xi_{1/3}$ and $L - \pi^2/9 = 2\xi_{2/3}$. Since μ is invariant under rational affine maps, this proves Theorem 1.2. □

Remark 11.4. In the formal deduction the nonvanishing is stated for prime intervals in the class 1 modulo 6 with the bad primes $\ell^2 \mid q$, because this is the general interface of the measure theorem; for the common-node minors of Theorem 11.2 it holds at every degree, so this interface imposes only the numerical condition $D - \Lambda < 1$, satisfied with $\eta = 0.63989$. It plays no role in the argument above.

12. The numbers $\psi'(\frac{a}{5}) - \zeta(2)$

The shifts $s = \frac{a}{5}$, $a \in \{1,2,3,4\}$, give four further numbers by the method of Section 11. Here the trigamma values are not combinations of $\zeta(2)$ and a single Dirichlet L-value, and we work with them directly:

$$\psi'(s) - \zeta(2) = \sum_{n \geq 0} \frac{1}{(n+s)^2} - \frac{\pi^2}{6}.$$

Write $c_a = \psi'(\frac{a}{5}) - \zeta(2)$. The multiplication formula gives $\sum_{a=1}^{5} \psi'(\frac{a}{5}) = 25\zeta(2)$, so $c_1+c_2+c_3+c_4 = \frac{10\pi^2}{3}$, and the reflection formula $\psi'(s)+\psi'(1-s) = \pi^2/\sin^2 \pi s$ gives

$$c_1 + c_4 = \frac{25+6\sqrt{5}}{15}\,\pi^2, \qquad c_2 + c_3 = \frac{25-6\sqrt{5}}{15}\,\pi^2.$$

A numerical search (PSLQ, 60 digits) finds no rational linear relation among 1 and any two of the c_a, so we treat Theorem 1.3 as four separate statements.

12.1. **Frames and kernels.** Let $s = \frac{a}{5}$, $u = 5 - a$, and put

$$F_s(z) = \psi(z+1) - \psi(z+s), \qquad \omega = \tfrac{1}{25},$$
$$\xi_s := \omega F_s'(0) = -\frac{1}{25}(\psi'(s) - \zeta(2)).$$

Then $k - 1 + s = (5k-u)/5$, and from $\psi(w+1) - \psi(w) = 1/w$,

$$F_s(k) - F_s(k-1) = \frac{1}{k} - \frac{5}{5k-u} = -\frac{u}{k(5k-u)},$$
$$F_s'(k) - F_s'(k-1) = -\frac{1}{k^2} + \frac{25}{(5k-u)^2}.$$

Telescoping gives the identity (2.2) with $\omega \mathsf{L}_{F_s}[R,Q] = M[R,Q] + \xi_s E_{R,Q}$ and the symmetric rational kernel

$$M[a',b'] = -\frac{u}{25(b'-a')} \sum_{k=a'+1}^{b'} \frac{1}{k(5k-u)} \quad (a' < b'),$$
$$M[a',a'] = -\frac{1}{25} \sum_{k=1}^{a'} \Big(\frac{1}{k^2} - \frac{25}{(5k-u)^2} \Big).$$

The geometries, the members P_J ($J = C$ and $J = C \setminus \{c\}$) and the admissibility condition (11.3), $r \leq 8m \leq r+9$, are those of Section 11.

12.2. **Content.**

Theorem 12.1 (content)**.** *Let $s = \frac{a}{5}$ with $a \in \{1,2,3,4\}$ and let r, m satisfy (11.3). For every $J \subseteq C$ with $|C \setminus J| \leq 1$ there is a positive integer $d_{r,m}$, independent of J, with $d_{r,m} P_J \in \mathbb{Z}[x]$ and*

$$\log d_{r,m} \leq (3.675334 + o(1))\, r^2,$$

uniformly in m.

The denominators of M divide $k(5k-u)$ and $b'-a'$. For a prime $p \geq 7$ the valuation analysis of §4 and Section 11 applies with $\Phi(k) = k(5k-u)$. For a prime power $q = p^f$ the set $\{k \geq 1 : q \mid \Phi(k)\}$ consists of the classes 0 and k_q, where $5k_q = u + j_q q$ with $j_q \in \{1,2,3,4\}$. So k_q/q is, up to $O(1/q)$, one of $\kappa \in \{\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\}$, determined by the class of q modulo 5, and by the prime number theorem in arithmetic progressions each value of κ occurs for primes of density $\frac{1}{4}$. Proposition 4.3 bounds the assignment value by an explicit function of r, m, q and k_q, with no new dual lemma. With $\theta = q/r$, the bound for $\theta \leq \frac{1}{4}$ is that of Section 11, with integral $\frac{1}{2} + \frac{1}{32} + \frac{\log 2}{32}$. For $\theta > \frac{1}{4}$ it is piecewise linear; for $\kappa = \frac{1}{5}$ the pieces are

$$\tfrac{17}{8} \text{ on } (\tfrac{1}{4}, \tfrac{15}{16}], \quad \tfrac{13}{4} - \tfrac{6}{5}\theta \text{ on } (\tfrac{15}{16}, 1], \quad \tfrac{7}{2} - \tfrac{7}{5}\theta \text{ on } (1, \tfrac{5}{4}],$$
$$\tfrac{9}{4} - \tfrac{2}{5}\theta \text{ on } (\tfrac{5}{4}, 5], \quad \tfrac{5}{4} - \tfrac{1}{5}\theta \text{ on } (5, \tfrac{25}{4}],$$

and the integrals over $\theta > \frac{1}{4}$ are

$$\kappa = \tfrac{1}{5} : \ \tfrac{7653}{1280}, \qquad \kappa = \tfrac{2}{5} : \ \tfrac{2637}{896}, \qquad \kappa = \tfrac{3}{5} : \ \tfrac{6149}{3072}, \qquad \kappa = \tfrac{4}{5} : \ \tfrac{451}{288}.$$

Averaging over the four classes,

$$\begin{aligned} \Lambda_{\mathrm{trop}} &= \frac{1}{4}\Big(\frac{7653}{1280} + \frac{2637}{896} + \frac{6149}{3072} + \frac{451}{288}\Big) + \frac{1}{2} + \frac{1}{32} + \frac{\log 2}{32} \\ &= \frac{4714081}{1290240} + \frac{\log 2}{32} = 3.67530\ldots, \end{aligned}$$

the same for all four values of a. (The same computation gives $\frac{16433}{5120} + \frac{\log 2}{32}$ for the thirds and $\frac{12527}{3360} + \frac{\log 2}{32}$ for the quarters.) The primes 2, 3 and 5, the prime powers with $f \geq 2$ and the finitely many exceptional primes contribute $o(r^2)$. In the formal proof the piecewise linear majorants are rational, with a small additive allowance for the rounding of m and of the residue classes, and their total is $\Lambda = \frac{195975263767}{53321748480} = 3.675334\ldots$; this is the value used below.

12.3. **Nonvanishing and decay.** For these frames

$$\delta_s(z) = \frac{1}{25}\Big(\frac{1}{z+1} - \frac{1}{z+s}\Big),$$

and the simple pole at $z = -1$ is again isolated in its class modulo $\mathbb{Z}$, because $-s \notin -1 + \mathbb{Z}$. The proof of Theorem 11.2 therefore applies word for word: for every $r \geq 1$, $m \geq 0$ and $x \in \mathbb{Q}$ some P_J with $|C \setminus J| \leq 1$ has $P_J(x) \neq 0$.

The decay estimate depends on the offset s only through the $o(1)$ terms (Section 11). Here the content is close to that of $G \pm 5\pi^2/48$, and the certificate of §9.4 applies unchanged: for admissible geometries, every J with $|C \setminus J| \leq 1$ and real t with $|t| \leq e^{-cr}$,

$$|P_J(\xi_s + t)| \leq \exp\big(-(D - o(1))r^2\big),$$
$$D = \tfrac{390557}{100000} = 3.90557, \qquad c = \tfrac{18386549}{2500000}.$$

12.4. **Proof of Theorem 1.3.** With

$$\eta = D - \Lambda = 3.90557 - 3.675334\ldots = 0.230235\ldots,$$
$$\frac{c}{\eta} = \frac{7.3546196}{0.230235\ldots} = 31.9439\ldots < 31.95,$$

the deduction is the one in the proof of Theorem 1.2, using Theorem 12.1, the nonvanishing above and the decay estimate. It gives $\mu(\xi_s) \le c/\eta < 31.95$, and $\psi'(\frac{a}{5}) - \zeta(2) = -25\xi_{a/5}$. □

13. Limits of the method

The exponents are determined by three constants: the content Λ, the decay D and the window constant c, through $c/(D-\Lambda)$. For λ_-, $\partial\mu/\partial D = -c/\eta^2 \approx -158$, so the third decimal of D matters. The rank deficits of §4, the uniform rescaled-core deficit of Section 5 and the next-level deficit of Section 6 determine Λ; the certificate of §9.4 determines D and c. The free choice of ℓ (Theorem 7.1 for all S in a range) removes the term $c\,g\,s$ of the deduction, which would otherwise be about 5.1.

For $L(2,\chi_{-3})$, characteristic-zero selection of a single common-node deletion allows the quarter border of Section 8; with the certificate there the ratio is $14.2095662593\ldots$. A dual measure, checked with 192-bit ball arithmetic, shows that no certificate of the same form, with the same α, slope, segment, ε and ρ, gives an exponent below 14.18. At this geometry, reducing the number of deleted nodes further would not change the leading decay or content constants. Further progress requires a smaller content constant, through new rank deficits, or a larger decay constant, through a better treatment of the energy problem; weak duality (Remark 9.4) measures how much a certificate of the present form can still gain. For $L(2,\chi_{-3}) \pm \frac{2}{3}\zeta(2)$ the certificate of Section 11 gives $10.6078\ldots$. A dual measure, checked with 192-bit ball arithmetic, shows that no certificate of the same form, with the same α, slope, segment and $\rho \ge 1/4096$, gives an exponent below 10.593, so a further gain needs a different geometry or a smaller content constant.

14. Formalization

All results of this paper are formalized in Lean 4 [11] with Mathlib [45]. The development is available at

`https://github.com/slitsey/catalan-irrationality-measure`

and combines analytic and arithmetic proofs with generated rational data and kernel checks for the dual certificates of Section 4 and the decay certificates of Section 9; numerical computations alone are not used as proof. The constants are written out as series, so the main statements use no definitions of the development: G is $\sum_j (-1)^j/(2j+1)^2$ and $L(2,\chi_{-3})$ is $\sum_k \big((3k+1)^{-2} - (3k+2)^{-2}\big)$. The main statements are stated in `ChallengeHybrid.lean` and proved in `SolutionHybrid.lean` (namespace `CatalanHybridChallenge`):

- `lam_hybrid_measure_3244`: for every $e > 0$, all sufficiently large $q \in \mathbb{N}$ and all $p \in \mathbb{Z}$ coprime to q, $q^{-(32.44+e)} < |G - 5\pi^2/48 - p/q|$ (Theorem 1.1);
- `mir_hybrid_measure_3244`: the same for $G + 5\pi^2/48$;
- `chi3_plus_zeta2_measure_1061`, `chi3_minus_zeta2_measure_1061`: the same for $L(2,\chi_{-3}) + \pi^2/9$ and $L(2,\chi_{-3}) - \pi^2/9$ with exponent 10.61 (Theorem 1.2); they are stated in `CatalanFrames/Thirds/StatementsT.lean` (namespace `CatalanFrames.Thirds`), with the content `thirdsSharpPrice` and the constants $D = \frac{77423}{20000}$, $c = \frac{169697}{25000}$ entering through `chi3_plus_zeta2_measure_of_decay` and the check `thirds_exponent_checkT`;
- `chi3_hybrid_measure_1421`: the same for $L(2,\chi_{-3})$ with exponent 14.21; it is `CatalanFrames.P6Hybrid.QuarterContour.Cert1421.chi3_measure`, with the decay certificate in `CatalanCore/Cert/QuarterContour` and `CatalanCore/Cert/SigmaQuarterContour`;
- `fifths1_hybrid_measure_3195`, ..., `fifths4_hybrid_measure_3195`: the same for $\psi'(\frac{a}{5}) - \pi^2/6$, $a = 1,\dots,4$, with exponent 31.95 (Theorem 1.3); they are `CatalanFrames.Fifths.fifths1_measure_3195`, ..., in `CatalanFrames/Fifths/Statements.lean`, with the content bound `fifths1_hasContent` and the decay `fifths1_minorDecayS`;
- `Quarter.decay`, `Quarter.Periodic.hasContent` and `Quarter.hasNVPin`: the quarter-border decay, content and characteristic-zero selection interfaces;
- `thirds_hasNVPinC`, `thirdsAc_minorDecayT` and `thirdsAc_hasContent_sharp`, with the `thirdsBc` versions: Theorems 11.2, 11.3 and 11.1;
- `longCell_rowLocal`, `kind3_rowLocal` and `lam_both_eps_merged`, `mir_both_eps_merged`: the next-level deficit of Section 6 on a long cell, its binding to the prime profile and the resulting content bound;
- `lam_hybrid_decayS`: Theorem 9.1 for λ_- with the certificate of §9.4; the decay estimate in `CatalanCore/Decay` takes the ray slope as a parameter;
- `nv_hyb_nonvanishing_gen`: Theorem 7.1 and its analogues in §7.7;
- `hyb_primal_false` and `primal_pair`: the end of the nonvanishing chain and Lemma 7.4;
- `profile_both_eps`: Theorem 4.1 for λ_- (the case of $L(2,\chi_{-3})$ is `Quarter.Periodic.hasContent`, above).

The solution has no `sorry` and no `native_decide`, and the command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The file `comparator-hybrid.json` is a configuration for `lake comparator` stating that `SolutionHybrid` proves the statements of `ChallengeHybrid` with these axioms only. We have not run Comparator, which requires Linux; the axiom check above was done with `#print axioms`.

The prime number theorem in arithmetic progressions (moduli 4 and 6) enters the content asymptotics and the selection of the prime ℓ. The formal source for it is the PNT+ project [46], through OpenAI's workspace and compatibility additions, whose Wiener–Ikehara theorem we use.
The following file groups implement the indicated parts of the proof.

`CatalanCore/Rect`
Geometries, the matrices $\Phi_{\mathcal{G}}$, Lemma 3.3, pair families and admissible relations.

`CatalanCore/Arith/Hybrid`
Gauss valuations, Lemma 4.2, the level majorant, Proposition 4.3, prime sums over linear pieces, the rank-deficit interface (Lemma 4.5) and bordering.

`CatalanCore/Arith/Hybrid/ShiftNonclean`, `CatalanFrames/HybCommon/DeficitProfile`
Proposition 6.2 and Theorem 6.1: the weighted congruence, the two-level determinant bound and the merged profile.

`CatalanFrames/Lambda/Hyb*`, `CatalanFrames/LambdaMirrorHybrid`, `CatalanFrames/P6Hybrid`
Theorem 4.1 and Theorem 4.6 for the three constants.

`CatalanFrames/LambdaHybrid/NV`
Proposition 7.3 and its three-level analogue, Lemma 7.4 (`Primal`), Theorem 7.1 (`FinalNV`).

`CatalanCore/Star/Leonard`
Proposition 7.8, Proposition 7.9, (K1)–(K5), the diagonal and Darboux steps, the positivity argument, Lemma 7.5, Theorem 7.7 and the specialisation $\theta \to 0$.

`CatalanCore/Decay/Rect`, `CatalanCore/Cert/SigmaHybS`, `CatalanCore/Cert/SigmaP6H`, `CatalanCore/Cert/SigmaThirds`
The contour identity, the two-family Andréief bound, Lemma 9.2, node removal, and the certificates of Proposition 9.3, Section 8 and Theorem 11.3.

`CatalanFrames/Thirds`
The frames `thirdsAc` and `thirdsBc` ($s = 1/3, 2/3$), the content tables and hybrid content, the characteristic-zero nonvanishing, the decay instance and the final statements of Section 11; the theorems are checked with `#print axioms` in each module.
par`CatalanCore/Measure`, `CatalanFrames/*/Assembly*`

The deduction of Section 10.

The proofs were checked with Lean `v4.35.0-rc2` and the matching Mathlib release. The vendored files from OpenAI's repository and from the PNT+ project were converted to Lean's module system and ported to the current

toolchain; the repository's README and provenance documents describe the changes.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author, who posed the problems, directed the work and is responsible for this paper: Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. The underlying irrationality proof for G and its Lean formalization are OpenAI's [36]. We use Lean 4 [11], Mathlib [45], the PNT+ project [46], and Arb [21] (through `python-flint`) for interval arithmetic in generating certificate data, which Lean re-checks. The formal verification in Lean is intended as an independent check of every step of the argument above.

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References

1. George E. Andrews, Richard Askey, and Ranjan Roy, *Special functions*, Encyclopedia of Mathematics and its Applications, vol. 71, Cambridge University Press, Cambridge, 1999.
2. Roger Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.
3. Keith Ball and Tanguy Rivoal, *Irrationalité d'une infinité de valeurs de la fonction zêta aux entiers impairs*, Invent. Math. **146** (2001), no. 1, 193–207.
4. Frits Beukers, *A note on the irrationality of $\zeta(2)$ and $\zeta(3)$*, Bull. London Math. Soc. **11** (1979), no. 3, 268–272.
5. Francis Brown, *Mellin transforms, transfinite diameter and rational approximations of integrals*, 2026, Preprint, 22 April 2026, arXiv:2604.20741.
6. Frank Calegari, Vesselin Dimitrov, and Yunqing Tang, *The linear independence of 1, $\zeta(2)$, and $L(2,\chi_{-3})$*, 2024, arXiv:2408.15403.
7. ———, *Arithmetic holonomy bounds and effective Diophantine approximation*, 2025, Theorem 5.1 gives $\mu(L(2,\chi_{-3})) \le 24781$, arXiv:2510.04156.
8. Marcos Costa Santos Carreira, *Determinant constructions for the Dirichlet beta value at four*, Preprint, 9 October 2026, doi:10.5281/zenodo.23326647, 2026.
9. ———, *General designs and an irrationality exponent for Catalan's constant*, Preprint, 9 October 2026, revised 21 October 2026, doi:10.5281/zenodo.23532239; sources at `https://github.com/MarcosCarreira/catalan-designs`, 2026.
10. Harold Davenport, *Multiplicative number theory*, third ed., Graduate Texts in Mathematics, vol. 74, Springer, New York, 2000, Revised and with a preface by H. L. Montgomery.
11. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
12. Payman Eskandari, *Improved rational approximations to Catalan's constant*, 2026, Preprint, 22 September 2026, arXiv:2609.26354.
13. Payman Eskandari, Kumar Murty, and Yusuke Nemoto, *Mixed motives and linear forms in the Catalan constant*, 2025, Preprint, October 2025 (v3: 29 August 2026), arXiv:2510.20648.

14. Aabir Fauzan, $\zeta(5)$ *is irrational*, Preprint v1, September 17, 2026, Zenodo, doi:10.5281/zenodo.22826419, 2026.
15. Peter J. Forrester, *Meet Andréief, Bordeaux 1886, and Andreev, Kharkov 1882–83*, 2018, arXiv:1806.10411.
16. David Niedbala Giraudin, *A new upper bound for the irrationality exponent of* $\zeta(2)$, 2026, Preprint, 2 October 2026, arXiv:2610.02912.
17. ———, *A new upper bound for the irrationality exponent of* $\zeta(3)$, 2026, Preprint, 22 September 2026, arXiv:2609.26980.
18. F. Alberto Grünbaum and Luc Haine, *Bispectral Darboux transformations: an extension of the Krall polynomials*, Int. Math. Res. Not. (1997), no. 8, 359–392.
19. F. Alberto Grünbaum and Milen Yakimov, *Discrete bispectral Darboux transformations from Jacobi operators*, Pacific J. Math. **204** (2002), no. 2, 395–431, arXiv:math/0012191.
20. R. K. Jha, *A finite irrationality measure for Catalan's constant*, Preprint v1, October 8, 2026, doi:10.5281/zenodo.23237269; Lean sources at `https://github.com/rohitkrjha/catalan-measure`, 2026.
21. Fredrik Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292, arXiv:1611.02831.
22. Roelof Koekoek, Peter A. Lesky, and René F. Swarttouw, *Hypergeometric orthogonal polynomials and their q-analogues*, Springer Monographs in Mathematics, Springer, Berlin, 2010.
23. Christian Krattenthaler, *Advanced determinant calculus*, Sém. Lothar. Combin. **42** (1999), Art. B42q, 67 pp., arXiv:math/9902004.
24. ———, *Advanced determinant calculus: a complement*, Linear Algebra Appl. **411** (2005), 68–166, arXiv:math/0503507.
25. Christian Krattenthaler and Tanguy Rivoal, *On a linear form for Catalan's constant*, South East Asian J. Math. Math. Sci. **6** (2008), no. 2, 3–15, arXiv:0810.1927.
26. Christian Krattenthaler and Wadim Zudilin, *Hypergeometry inspired by irrationality questions*, Kyushu J. Math. **73** (2019), no. 1, 189–203, arXiv:1802.08856.
27. Li Lai, *The linear independence of* 1, $\zeta(3)$, *and* $\zeta(4)$, 2026, Preprint, 5 October 2026 (v2: 15 October 2026), arXiv:2610.06838.
28. Elyasheev Leibtag, Hila Barkan, Rotem Kalisch, Shachar Weinbaum, Michael Shalyt, and Ido Kaminer, *New directions in irrationality*, 2026, Preprint, 20 October 2026, arXiv:2610.27799; supplement and code at `https://github.com/RamanujanMachine/new-directions-in-irrationality`.
29. Sean Litsey, *An irrationality measure for Catalan's constant*, Hexagon preprint hexagon:2610.00187v4, `https://hexagonmath.org/2610.00187v4`, 2026.
30. Jingwen Liu, Kai Jiang, and Pingwen Zhang, *Irrationality exponents of logarithms of positive rational numbers*, 2026, Preprint, 7 October 2026, arXiv:2610.10192.
31. Junqi Liu, Jujian Zhang, and Lihong Zhi, *A formal proof of the irrationality of* $\zeta(3)$ *in Lean 4*, 2025, arXiv:2503.07625.
32. Karl Löwner, *Über monotone Matrixfunktionen*, Math. Z. **38** (1934), 177–216.
33. Raffaele Marcovecchio, *The Rhin–Viola method for* $\log 2$, Acta Arith. **139** (2009), no. 2, 147–184.
34. Yu. V. Nesterenko, *On the linear independence of numbers*, Vestnik Moskov. Univ. Ser. I Mat. Mekh. (1985), no. 1, 46–49, In Russian; English translation: Moscow Univ. Math. Bull. 40 (1985), no. 1, 69–74.
35. E. M. Nikishin and V. N. Sorokin, *Rational approximations and orthogonality*, Translations of Mathematical Monographs, vol. 92, American Mathematical Society, Providence, RI, 1991.
36. OpenAI, *Catalan's constant is irrational*, Preprint dated September 24, 2026, released October 6, 2026, with a Lean formalization, available at the original paper PDF and the Lean formalization (repository commit `fd4aeeb`, October 7, 2026), 2026.

37. ———, *The irrationality exponent of π is 2*, Preprint dated September 24, 2026, released October 6, 2026, https://github.com/openai/math/tree/main/preprints/The-irrationality-exponent-of-pi-is-2-September-24-2026, 2026.
38. Thomas Prellberg, *Three irrationality results for the dilogarithm: Rhin–Viola and Viola–Zudilin constructions at $-1/4$, $1/5$, and $-1/3$*, 2026, Preprint, 7 September 2026, arXiv:2609.06895.
39. Georges Rhin and Carlo Viola, *On a permutation group related to $\zeta(2)$*, Acta Arith. **77** (1996), no. 1, 23–56.
40. ———, *The group structure for $\zeta(3)$*, Acta Arith. **97** (2001), no. 3, 269–293.
41. Tanguy Rivoal and Wadim Zudilin, *Diophantine properties of numbers related to Catalan's constant*, Math. Ann. **326** (2003), no. 4, 705–721.
42. Edward B. Saff and Vilmos Totik, *Logarithmic potentials with external fields*, Grundlehren der mathematischen Wissenschaften, vol. 316, Springer, Berlin, 1997.
43. Vyacheslav Spiridonov and Alexei Zhedanov, *Spectral transformations, self-similar reductions and orthogonal polynomials*, J. Phys. A **30** (1997), no. 21, 7621–7637.
44. Zhi-Wei Sun, *Catalan's constant is irrational*, 2026, arXiv:2609.04176.
45. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 2020, pp. 367–381.
46. The PrimeNumberTheoremAnd contributors, *The PNT+ project: a Lean formalization of the prime number theorem and related results*, https://github.com/AlexKontorovich/PrimeNumberTheoremAnd, The vendored compatibility additions used here are recorded in the repository's `NOTICE` and `PORTING.md`.
47. Dane Wachs, *A note on a recent claimed proof of the irrationality of Catalan's constant*, 2026, arXiv:2609.22339.
48. Doron Zeilberger and Wadim Zudilin, *The irrationality measure of π is at most* 7.103205334137..., Mosc. J. Comb. Number Theory **9** (2020), no. 4, 407–419, arXiv:1912.06345.
49. Alexei Zhedanov, *Rational spectral transformations and orthogonal polynomials*, J. Comput. Appl. Math. **85** (1997), no. 1, 67–86.
50. Wadim Zudilin, *A few remarks on linear forms involving Catalan's constant*, Chebyshevskii Sb. **3** (2002), no. 2(4), 60–70, In Russian, arXiv:math/0210423.
51. ———, *An Apéry-like difference equation for Catalan's constant*, Electron. J. Combin. **10** (2003), Research Paper 14, arXiv:math/0201024.
52. ———, *A determinantal approach to irrationality*, Constr. Approx. **45** (2017), no. 2, 301–310, arXiv:1507.05697.
53. ———, *Arithmetic of Catalan's constant and its relatives*, Abh. Math. Semin. Univ. Hambg. **89** (2019), no. 1, 45–53, arXiv:1804.09922.

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